

Exact recovery in the double sparse model: Sufficient and necessary signal conditions[†]

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Abstract: The double sparse linear model, which has both group-wise and element-wise sparsity in regression coefficients, has attracted lots of attention recently. This paper establishes the sufficient and necessary relationship between the exact support recovery and the optimal minimum signal conditions in the double sparse model. Specifically, sharply under the proposed signal conditions, a two-stage double sparse iterative hard thresholding procedure achieves exact support recovery with a suitably chosen threshold parameter. Additionally, this procedure maintains asymptotic normality, aligning with the OLS estimator given true support, thereby holding the oracle properties. Conversely, we prove that no method can achieve exact support recovery if these signal conditions are violated. This fills a critical gap in the minimax optimality theory on support recovery of the double sparse model. Numerical experiments are provided to support our theoretical findings.

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1. Introduction

Consider the double sparse linear model under simultaneous group sparsity and element sparsity:

$$Y = X\beta^* + \sigma\xi \in \mathbb{R}^n, \quad (1)$$

where $Y \in \mathbb{R}^n$ is the response variable, $X \in \mathbb{R}^{n \times p}$ is the design matrix and $\beta^* \in \mathbb{R}^p$ is the coefficient vector. The noise level σ and the 1-subgaussian random vector $\xi \in \mathbb{R}^n$ (independent of X) together constitute the random term. Assume the p covariates can be partitioned into m non-overlapping groups $\{G_j\}_{j=1}^m$ with equal group size d (for groups with different sizes, we take $d = \max\{|G_j|\}_{j=1}^m$ and the results are not affected), such that $\sum_{j=1}^m |G_j| = m \times d = p$. Following the definition in [11], we assume that the coefficient vector β^* belongs to the double sparse parameter space:

$$\beta^* \in \Theta(s, s_0) := \left\{ \beta \in \mathbb{R}^p \left| \sum_{j=1}^m \mathbf{1}(\beta_{G_j} \neq \mathbf{0}_d) \leq s, \|\beta\|_0 \leq s s_0 \right. \right\}, \quad (2)$$

where $\|\beta\|_0$ denotes the number of nonzero components of β , and $\mathbf{1}(\cdot)$ denotes the indicator function. The double sparse structure in (2) implies that the number of support groups does not exceed s , and the total number of support elements does not exceed $s s_0$. And we claim that β is (s, s_0) -sparse if $\beta \in \Theta(s, s_0)$.

This double sparsity is particularly valuable in applications such as change-point detection [45, 36], three-dimensional image formation [42], multi-task learning [2], among many others. Theoretically, most existing studies [11, 23, 46, 22] focused on the minimax optimal estimation of the signal vector β^* based on ℓ_2 risk. However, the performance of support recovery (i.e., variable selection) and the asymptotic distribution in the double sparse model remain insufficiently explored. This paper addresses the theoretical gap by thoroughly analyzing the influence of signal strength on exact support recovery and statistical inference in model (1). From a non-asymptotic perspective, we establish the necessary and sufficient minimum signal conditions for achieving exact recovery in the double sparse space.

1.1. Related work

Double sparse model The double sparse structure is of practical significance, attracting extensive research in both algorithmic development [24, 6] and theoretical exploration [11, 23, 22]. Theoretically, [11, 23] have established minimax lower bounds for signal estimation in the double sparse model (1) as

$$\inf_{\hat{\beta}} \sup_{\beta^* \in \Theta(s, s_0)} \mathbf{E} \left(\|\hat{\beta} - \beta^*\|_2^2 \right) \gtrsim \frac{\sigma^2}{n} (s \log(m/s) + s s_0 \log(d/s_0)), \quad (3)$$

where the infimum is taken over all possible estimators $\hat{\beta}$ based on $(Y, X) \in \mathbb{R}^n \times \mathbb{R}^{n \times p}$, and $\gtrsim$ denotes the inequality up to an absolute constant. [11, 22] provided theoretical analyses of the Sparse Group Lasso (SGLasso), which is introduced by [35]. This algorithm combines the Lasso penalty [37] with the group Lasso penalty [43], integrating both element-wise and group-wise sparsity. [46] proposed a tuning-free Double Sparse Iterative Hard Thresholding (DSIHT) algorithm and showed that the obtained estimator is minimax optimal. However, as [13, 14, 49] pointed out, under proper conditions, a good estimator should possess exact recovery and asymptotic normality, i.e.,

$$\text{supp}(\hat{\beta}) := \text{supp}(\beta^*), \quad \sqrt{n} (\hat{\beta}_{S^*} - \beta_{S^*}^*) \xrightarrow{d} N(0, \Sigma^*), \quad (4)$$

where Σ^* is the covariance matrix on the true support. In general, these two properties in (4) are called the oracle properties. To the best of our knowledge, the oracle properties of the double sparse model are currently absent in existing works [11, 23, 46, 22], which necessitates further analysis on these properties. In the element-wise sparse linear model (assuming only $\|\beta^*\|_0 \leq s$), [30] proposed an IHT-type algorithm that ensures minimax optimality and achieves exact support recovery. This finding motivates us to investigate whether the IHT-type algorithm possesses oracle properties in the double sparse setting.

Support recovery The minimax optimality of exact recovery, i.e., variable selection consistency, is a heatedly discussed topic [39, 41, 10, 9, 8]. Specifically, it corresponds to a minimax separate rate μ , where there exist some algorithms that can achieve exact support recovery if the minimum signal strength $\min_{j \in S^*} |\beta_j^*| > C_1 \mu$, and no algorithm can ensure such selection consistency if $\min_{j \in S^*} |\beta_j^*| < C_2 \mu$, with $C_1 \geq C_2 > 0$ being two constants. In a p -dimensional Gaussian sequence model $X \sim N(\beta^*, \sigma^2 I_n)$ with only element-wise sparsity $\|\beta^*\|_0 \leq s$, [9] identified the minimax separate rate for exact recovery as $\sqrt{2\sigma^2 \log p}$. Recent studies [16, 3, 1] also explored this issue, emphasizing the necessity of minimum signal conditions for exact recovery. Additionally, [29] revealed how support recovery enhances signal estimation from the perspective of the minimax estimation lower bound, highlighting the interaction between these two. However, these studies concentrate on element-wise sparsity, and their conclusions cannot be directly extended to the double sparse space (2). Therefore, a detailed investigation into the exact recovery for the double sparse model is necessary.

In summary, based on the existing studies, we propose the following questions:

In the double sparse model (1), can the IHT-type algorithm achieve exact support recovery? If so, are the minimum signal conditions required for this recovery minimax optimal? Can these findings facilitate the oracle properties (4)?

This paper presents a theoretical analysis of the aforementioned questions and provides affirmative answers.

1.2. Main contribution

This paper establishes the sufficient and necessary relationship between the exact support recovery and the optimal minimum signal conditions in the double sparse model. We demonstrate the oracle properties of the double sparse IHT procedure, further enhancing its theoretical foundation. Specifically, the main contributions of this paper are summarized as follows:

1. Theoretically, we show that the double sparse iterative hard thresholding procedure possesses oracle properties (4), guaranteeing that its output achieves both exact support recovery and oracle asymptotic normality, under our proposed minimum signal conditions. This result confirms the sufficiency of these signal conditions for achieving exact recovery.
2. We analyze the minimax lower bounds for selection error based on Hamming risk. Results show that no method can achieve exact support recovery if the proposed minimum signal conditions are violated. This result confirms the necessity of these signal conditions and highlights that the double sparse IHT procedure achieves minimax rate-optimality in support recovery.

The innovation of this paper lies in analyzing how signal strength influences the exact recovery and statistical inference of the double sparse linear model (1). From a non-asymptotic framework, we show that a two-stage DSIHT estimator converges to the oracle estimator under rate-optimal signal conditions, emphasizing the theoretical superiority over convex penalized estimators such as the sparse group Lasso. Table 1 provides a comparison between double sparse IHT and sparse group Lasso algorithms.

Additionally, compared to the element-wise sparse model or the group-wise sparse model, the double sparse model provides richer information about the structure of β^* , thus implying the potential for more accurate outcomes (i.e., smaller sample complexity [11]). To achieve exact support recovery, we need to analyze the signal conditions from both the group and

Table 1

Comparison of Sparse group lasso and Double sparse IHT algorithms in the double sparse model

	Minimax lower bound	Sparse group Lasso	Double sparse IHT
Signal estimation	Equation (3), Cai et al. [11]	Rate-optimal, Cai et al. [11]	Rate-optimal, Zhang et al. [22]
Exact recovery	Theorems 3.2 and 3.3, ours	Unknown	Theorem 2.2, ours
Asymptotic Normality	NA	De-sparsified normality, [11] (under more stringent conditions)	Theorem 2.3, ours Distributed as oracle estimator

element perspectives, examining how each signal condition reduces the estimation (and selection) error. This problem is more challenging than addressing either the element-wise [9] or group-wise [26, 8] cases individually.

1.3. Organization

The rest of the paper is organized as follows: Section 1 establishes the problem and notations used throughout the paper. Section 2 introduces the double sparse IHT procedure and proves its final estimator has oracle properties under suitable signal conditions. Section 3 establishes the minimax lower bound for support recovery based on Hamming loss in model (1), showing that the minimum signal conditions required in Section 2 are minimax rate-optimal. Section 4 presents numerical experiments to confirm our theoretical findings. Section 5 contains the conclusion and possible extensions of our study. Appendix A, B, C and D provide the proof of our results.

1.4. Notations and preliminaries

For the given sequences a_n and b_n , we say that $a_n = O(b_n)$ when $a_n \leq Cb_n$ for some constant $C > 0$, while $a_n = o(b_n)$ corresponds to $a_n/b_n \rightarrow 0$ as $n \rightarrow \infty$. We write $a_n \asymp b_n$ if $a_n = O(b_n)$ and $b_n = O(a_n)$. Let $[m]$ denote the set $\{1, 2, \dots, m\}$. Let $x \vee y = \max\{x, y\}$, and $x \wedge y = \min\{x, y\}$. For a vector β , let $\|\beta\|_2$ denote its Euclidean norm and $\|\beta\|_0$ denotes the number of its nonzero components. For a set S , let $|S|$ denote its cardinality. For a matrix A , let $\|A\|_2$ denote its spectral norm and $\|A\|_F$ denotes its Frobenius norm. Let $\mathbf{I}_p \in \mathbb{R}^{p \times p}$ denote the identity matrix, and $\mathbf{0}_d \in \mathbb{R}^d$ denote the d -dimensional zero vector. We use C_i to denote absolute constants, whose actual values vary from time to time.

We next introduce some more specific notations related to the double sparse model. We use the double index (i, j) to locate the i -th variable in the j -th group G_j for $i \in [d]$, $j \in [m]$ in the original parameter vector. Under the given group structure, each element's location in a vector corresponds to a unique location with the double index; therefore, we use these two notations interchangeably without further declaration. For a fixed vector $\beta \in \mathbb{R}^p$, we denote by $\text{supp}(\beta) := \{(i, j) \in [d] \times [m] : \beta_{ij} \neq 0\}$ the support set of β , and $G^*(\beta) := \{j \in [m] : \beta_{G_j} \neq \mathbf{0}_d\}$ the group index set of true support groups. We refer to the coefficient vector β as (s, s_0) -double sparse if $\beta \in \Theta(s, s_0)$. Denote by $\mathcal{S}(s, s_0) := \{\text{supp}(\beta) : \beta \in \Theta(s, s_0)\}$ the space consisting of all the support set of $\Theta(s, s_0)$, that is, if $\beta \in \Theta(s, s_0)$, we say the support of β belongs to $\mathcal{S}(s, s_0)$. Furthermore, we denote by $\beta_S \in \mathbb{R}^{|S|}$ the subvector of β indexed by the set S , where S can be any subset of the index space $[d] \times [m]$. Notably, our results can also be considered as $n \rightarrow \infty$ when other parameters, i.e., d, m, s , and s_0 , all depend on n in such a way that $d = d(n) \rightarrow \infty$. For brevity, the dependence of these parameters on n will be further omitted in the notation.

To facilitate our technical derivation, the design matrix is standardized as $\|X_{(ij)}\|_2 = \sqrt{n}$, where $X_{(ij)} \in \mathbb{R}^n$ denotes the corresponding observation vector of the variable (i, j) , for all $(i, j) \in [d] \times [m]$. Furthermore, we introduce a fundamental assumption for the design matrix X , termed the Double Sparse Restricted Isometry Property (DSRIP). This assumption is originally introduced in [23] and is an extension of the standard RIP [12] into the double sparse space.

Definition 1.1 (DSRIP condition). We say that $X \in \mathbb{R}^{n \times p}$ satisfies the Double Sparse Restricted Isometry Property $DSRIP(as, bs_0, \delta)$ with a constant $0 < \delta < 1$, if and only if

$$n(1 - \delta)\|u\|_2^2 \leq \|X_S u\|_2^2 \leq n(1 + \delta)\|u\|_2^2 \quad (5)$$

holds for all $S \in \mathcal{S}(as, bs_0)$ and $u \in \mathbb{R}^{|S|} \setminus \{\mathbf{0}_{|S|}\}$, where $a, b > 0$ are two constants, and $X_S \in \mathbb{R}^{n \times |S|}$ is the design matrix of the variables indexed by S .

The DSRIP condition is less strict compared to the RIP condition when considering the double sparsity. Specifically, taking $a = b = 1$, DSRIP requires an isometry property for all (s, s_0) -double sparse vectors. In contrast, RIP requires the satisfaction of all ss_0 -sparse vectors.

2. The double sparse IHT algorithm and oracle properties

The Iterative Hard Thresholding (IHT) algorithm is an effective method that plays a significant role in compressed sensing [5, 20, 44, 25]. This section illustrates the theoretical properties of the IHT-type algorithm regarding exact support recovery and asymptotic distribution in the double sparse model. First, Section 2.1 introduces the DSIHT Algorithm 1 proposed by [46], illustrating that it does not address the analysis of oracle properties. Section 2.2 then introduces a two-stage DSIHT Algorithm 2 that refines the output of Algorithm 1. We show that the final estimator from Algorithm 2 achieves a sharper estimation rate and exhibits oracle properties under suitable minimum signal conditions, thereby filling a theoretical gap in the double sparse regression model.

The analyses in this section are based on the fixed design matrix X satisfying the DSRIP condition (see Definition 1.1). For clarity, we set the coefficient vector $\beta^* \in \Theta(s, s_0)$ in model (1) as fixed. We denote its support group index set as $G^* = G^*(\beta^*)$ and its support set as $S^* = S^*(\beta^*)$.

2.1. One-stage algorithm: Minimax optimal estimation

First, we introduce the double sparse thresholding operator $\mathcal{T}_{\lambda, s_0} : \mathbb{R}^p \rightarrow \mathbb{R}^p$ proposed by [46]. This operator comprises the following two steps:

1. The element-wise operator:

$$\left\{ \mathcal{T}_{\lambda}^{(1)}(\beta) \right\}_{ij} = \beta_{ij} \cdot \mathbf{1}(|\beta_{ij}| \geq \lambda) \in \mathbb{R}. \quad (6)$$

2. The group-wise operator:

$$\left\{ \mathcal{T}_{\lambda, s_0}^{(2)}(\beta) \right\}_{G_j} = \beta_{G_j} \cdot \mathbf{1}\left(\|\beta_{G_j}\|_2^2 \geq s_0 \lambda^2\right) \in \mathbb{R}^d. \quad (7)$$

Then, the double sparse hard thresholding operator can be described as $\mathcal{T}_{\lambda, s_0} := \mathcal{T}_{\lambda, s_0}^{(2)} \circ \mathcal{T}_{\lambda}^{(1)}$. Specifically, for each group G_j , if $\sum_{i=1}^d \left\{ \hat{\beta}_{ij}^2 \mathbf{1}(|\hat{\beta}_{ij}| \geq \lambda) \right\} \geq s_0 \lambda^2$, group G_j will be estimated as a support group and $\{\mathcal{T}_{\lambda, s_0}(\hat{\beta})\}_{ij} = \hat{\beta}_{ij} \mathbf{1}(|\hat{\beta}_{ij}| \geq \lambda)$ for each $(i, j) \in G_j$. Conversely, if $\sum_{i=1}^d \left\{ \hat{\beta}_{ij}^2 \mathbf{1}(|\hat{\beta}_{ij}| \geq \lambda) \right\} < s_0 \lambda^2$, group G_j will be estimated as a non-support group, therefore $\{\mathcal{T}_{\lambda, s_0}(\hat{\beta})\}_{G_j} = \mathbf{0}_d$.

Algorithm 1 DSIHT (Double Sparse IHT)

Input: $X, Y, \{G_j\}_{j=1}^m, \kappa, \delta, \lambda_{(\infty)}, s_0, \sigma$

- 1: Initialize $t = 0, \lambda_{(0)} = \frac{\|X^\top Y/n\|_\infty + \sqrt{10\sigma^2(\log(dm))/n}}{\sqrt{2}\kappa}$ and $\hat{\beta}^0 = \mathbf{0}_p$
 - 2: **while** $t \leq \lceil \log(\lambda_{(\infty)}/\lambda_{(0)})/\log \kappa \rceil$, **do**
 - 3: $\hat{\beta}^{t+1} = \mathcal{T}_{\lambda_{(t)}, s_0} \left(\hat{\beta}^t + \frac{1}{n} X^\top (Y - X \hat{\beta}^t) \right)$
 - 4: $\lambda_{(t+1)} = \left(\kappa \lambda_{(t)} \right) \vee \lambda_{(\infty)}$
 - 5: $t = t + 1$
 - 6: **end while**
- Output:** $\hat{\beta}^t$
-

Leveraging the operator $\mathcal{T}_{\lambda, s_0}$, [46] introduced the Double Sparse Iterative Hard Thresholding (DSIHT) Algorithm 1, a gradient-descent-based procedure enforcing both group-wise and element-wise sparsity. At the t -th iteration, the threshold parameter follows $\lambda_{(t)} = \max\{\kappa^t \lambda_{(0)}, \lambda_{(\infty)}\}$, where $\kappa \in (0, 1)$ governs a decay from the initial value $\lambda_{(0)}$ to the floor $\lambda_{(\infty)}$. We next show that, by selecting $\lambda_{(\infty)}$ appropriately, this dynamic regularization strategy guarantees that the output of Algorithm 1 processes a double sparse structure, and its ℓ_2 error is effectively bounded.

For ease of display, we denote by $\{\hat{\beta}^t\}_{t \geq 0}$ the estimation sequence obtained from Algorithm 1. Define the element-wise decoder $\eta_{ij}(\beta) = \mathbf{1}(\beta_{ij} \neq 0)$ and the group-wise decoder $(\eta_G)_j(\beta) = \mathbf{1}(\beta_{G_j} \neq \mathbf{0}_d)$, for every $\beta \in \mathbb{R}^p$ with group structure $G_1, \dots, G_m$, $(i, j) \in [d] \times [m]$. Using these two decoders, we can characterize both the element-wise and group-wise support recovery errors (in terms of Hamming loss). Additionally, define

$$C_\lambda = C_\lambda(\kappa, \delta) := \sqrt{40} \times \frac{\kappa + (\sqrt{3} - 1)\delta}{\kappa - \delta}, \quad A = A(\kappa, \delta) := \frac{8\delta^2}{(\kappa - \delta)^2},$$

and

$$\Delta(s, s_0) := (1/s_0) \cdot \log(em/s) + \log(ed/s_0).$$

Theorem 2.1 (Estimation upper bound). *Assume that the design matrix X satisfies the condition DSRIP $\left((1 + 2A)s, \frac{1+4A}{1+2A}s_0, \delta \right)$ (see Definition 1.1) with $\delta \in (0, 1)$. Assume that $ss_0\Delta(s, s_0) = O(n)$ and $\kappa \in (\delta, 1)$. Then, by taking $\lambda_{(\infty)} = C_\lambda \sigma \cdot \sqrt{\Delta(s, s_0)/n}$, with a probability greater than $1 - \exp\{-(A \wedge 1)ss_0\Delta(s, s_0)/3\}$, for every $t \geq 0$ we have the following properties:*

1. The estimator $\hat{\beta}^t$ achieves sparse group selection as

$$\sum_{j=1}^m |(\eta_G)_j(\hat{\beta}^t) - (\eta_G)_j(\beta^*)| \leq (A + 1)s.$$

2. The estimator $\hat{\beta}^t$ achieves sparse element selection as

$$\sum_{j=1}^m \sum_{i=1}^d |\eta_{ij}(\hat{\beta}^t) - \eta_{ij}(\beta^*)| \leq (2A + 1)ss_0.$$

3. The estimator $\hat{\beta}^t$ achieves an upper bound as

$$\|\hat{\beta}^t - \beta^*\|_2 \leq \left(1 - \frac{\sqrt{10}}{C_\lambda}\right) \frac{2\sqrt{2}\kappa}{\kappa - \delta} \cdot \sqrt{ss_0\lambda_{(t)}}.$$

Although [46] established a result akin to Theorem 2.1, we restate it here for clarity and to support our subsequent analysis in Section 2.2: Theorem 2.1 proves that the support set of $\hat{\beta}^t$ remains double sparse, that is, $\hat{S}^t \in \mathcal{S}\left((1+A)s, \frac{1+2A}{1+A}s_0\right)$ holds for all $t \geq 0$, where the definition of $\mathcal{S}$ follows from Section 1.4. By terminating the iterations at $t = \lceil \log(\lambda_{(\infty)}/\lambda_{(0)})/\log \kappa \rceil$, the output of Algorithm 1 satisfies

$$\|\hat{\beta}^t - \beta^*\|_2^2 \leq 80 \cdot \frac{\left[\kappa + (2\sqrt{3} - 1)\delta\right]^2 \kappa^2}{(\kappa - \delta)^4} \cdot \frac{\sigma^2 [s \log(em/s) + ss_0 \log(ed/s_0)]}{n}, \quad (8)$$

which meets the rate in minimax lower bound (3), demonstrating that Algorithm 1 is rate-optimal. Moreover, Theorem 2.1 allows the DSRIP parameter δ to range freely over $(0, 1)$, and provides an explicit characterization of how both κ and δ influence the sparse pattern and estimation error of $\hat{\beta}^t$. This refined and quantitative insight into the dynamic regularization path goes beyond the earlier studies [23, 46].

2.2. Two-stage algorithm: Oracle properties

While the first-stage Algorithm 1 yields an estimator that achieves the minimax estimation rate, it tends to omit some true support variables and fall short in terms of estimation accuracy in practice. Furthermore, Theorem 2.1 does not ensure exact recovery results, nor does it provide the asymptotic distribution of its estimator. Next, we delve deeper into the analysis of the aforementioned issues.

The first goal is to obtain an estimator that achieves exact support recovery. To realize this, we propose the two-stage DSIHT Algorithm 2 with a fixed thresholding parameter μ , which will be clarified in Proposition 2.1. In the first stage, Algorithm 2 uses the output of Algorithm 1 as the initial input $\tilde{\beta}^0$, and then performs iterative updates in the second stage, producing the estimation sequence $\{\tilde{\beta}^t\}_{t \geq 0}$. The essence of Algorithm 2 lies in the utilization of the double sparse thresholding operator $\mathcal{T}_{\mu, s_0}$, which is derived similarly to (6) and (7).

For the analysis of exact recovery, let the oracle estimator $\tilde{\beta}^*$ be defined by

$$\tilde{\beta}_{S^*}^* = (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top Y, \quad \tilde{\beta}_{(S^*)^c}^* = \mathbf{0},$$

where $X_{S^*} \in \mathbb{R}^{n \times |S^*|}$ contains the columns of X indexed by the true support S^* , and $(S^*)^c$ is its complement. The next proposition quantifies the deviation $\tilde{\beta}^t - \tilde{\beta}^*$ produced by Algorithm 2 under suitable element-wise and group-wise minimum signal conditions. Recall $\Delta(1, s_0) = (1/s_0) \cdot \log(em) + \log(ed/s_0)$.

Algorithm 2 Two-stage DSIHT**Input:** $X, Y, \{G_j\}_{j=1}^m, \kappa, \delta, \lambda_{(\infty)}, \mu, s_0, \sigma$ 1: Initialize $t = 0$ 2: $\tilde{\beta}^0 = \text{DSIHT}(X, Y, \{G_j\}_{j=1}^m, \kappa, \delta, \lambda_{(\infty)}, s_0, \sigma)$ // first-stage initialization by Algorithm 13: **while** $t \leq C \log n$, **do**4: $\tilde{\beta}^{t+1} = \mathcal{T}_{\mu, s_0} \left(\tilde{\beta}^t + \frac{1}{n} X^\top (Y - X \tilde{\beta}^t) \right)$ // second-stage iteration with a fixed μ 5: $t = t + 1$ 6: **end while****Output:** $\tilde{\beta}^t$

Proposition 2.1 (Convergence to $\tilde{\beta}^*$). Assume that the design matrix X satisfies the condition $\text{DSRIP} \left((1 + 2A)s, \frac{1+4A}{1+2A}s_0, \delta \right)$ condition with $\delta \in (0, 1)$, and $ss_0\Delta(s, s_0) = O(n)$. We choose the fixed thresholding parameter μ as

$$\mu = \max \left\{ \frac{\kappa C_\lambda}{\delta}, \sqrt{40 + \frac{120\delta^2}{(1-\delta)^2}} \right\} \cdot \sqrt{\frac{\sigma^2}{n} \left\{ \frac{\log(em)}{s_0} + \log(esd) \right\}}, \quad (9)$$

and assume that both the element-wise and the group-wise minimum signal conditions

$$\begin{aligned} \min_{(i,j) \in S^*} |\beta_{ij}^*| &\geq \left(2 + \frac{\sqrt{6}\delta}{1-\delta} \right) \mu, \\ \min_{j \in G^*} \|\beta_{G_j}^*\|_2 &\geq \left(2 + \frac{\sqrt{6}\delta}{1-\delta} \right) \sqrt{s_0} \mu \end{aligned} \quad (10)$$

hold. Then, with probability greater than $1 - O \left(e^{-\frac{1}{3} \{ \Delta(1, s_0) + \log(ss_0) \}} \right)$, we have

$$\|\tilde{\beta}^t - \tilde{\beta}^*\|_2 \leq \left\{ \sqrt{5/6} + \left(1 - \sqrt{5/6} \right) \delta \right\}^t \cdot \|\tilde{\beta}^0 - \tilde{\beta}^*\|_2, \quad \text{for every } t \geq 0. \quad (11)$$

With the thresholding parameter μ in (9) and the minimum signal conditions (10), we deduce that the solution sequence $\{\tilde{\beta}^t\}_{t \geq 0}$ converges to the oracle estimator $\tilde{\beta}^*$ from a non-asymptotic perspective. This result quantifies both the optimization error and the computational efficiency of our algorithm, and serves as the key point for the exact support recovery and sharper error bound, as shown below.

Theorem 2.2 (Exact support recovery). Assume all conditions in Proposition 2.1 hold. Then, with probability greater than $1 - O \left(e^{-\frac{1}{3} \{ \Delta(1, s_0) + \log(ss_0) \}} \right)$, for all $t \geq C_\delta \log n$, $\tilde{\beta}^t$ achieves the exact support recovery at both the group-wise and element-wise levels

$$\sum_{j=1}^m |(\eta_G)_j(\tilde{\beta}^t) - (\eta_G)_j(\beta^*)| = 0, \quad \sum_{j=1}^m \sum_{i=1}^d |\eta_{ij}(\tilde{\beta}^t) - \eta_{ij}(\beta^*)| = 0.$$

With probability greater than $1 - \epsilon - O \left(e^{-\frac{1}{3} \{ \Delta(1, s_0) + \log(ss_0) \}} \right)$, it also achieves a sharper error bound as

$$\|\tilde{\beta}^t - \beta^*\|_2 \leq \sqrt{\frac{3\sigma^2}{1-\delta}} \left(\sqrt{\frac{ss_0}{n}} + \sqrt{\frac{\log(1/\epsilon)}{n}} \right). \quad (12)$$

Theorem 2.2 demonstrates that, under suitable element-wise and group-wise signal conditions, the output of Algorithm 2 can exactly recover the support set at both the group-wise and element-wise levels. Section 3 further demonstrates that our signal conditions (10) are necessary in terms of rate. Therefore, we show that the double sparse IHT procedure achieves minimax optimality in both signal estimation [46] and support set recovery.

Furthermore, under signal conditions (10), $\tilde{\beta}^t$ achieves the estimation rate $O\left(\sqrt{\frac{ss_0}{n}}\right)$ as if the support of β^* were known. This property is a key advantage of our IHT-based method and is generally difficult to achieve by convex procedures such as the sparse group Lasso [11].

Remark 2.1 (Initial estimator of Algorithm 2). Although Algorithm 2 uses the output of Algorithm 1 as the initial estimator $\tilde{\beta}^0$, in theory, it can be replaced with any estimator that achieves the minimax estimation rate (3). For example, the sparse group Lasso [11] or the sparse group slope [22] can serve as $\tilde{\beta}^0$, rendering our procedure broadly applicable. Therefore, the second-stage iteration in Algorithm 2 can be realized as a fast debiasing procedure: under signal condition (10), it removes the bias induced by regularization and ensures that $\tilde{\beta}^t$ converges to the oracle estimator.

Remark 2.2 (A delicate proof technique for support recovery). Exact recovery of the support set S^* conventionally relies on bounding the ℓ_∞ error $\|\hat{\beta} - \beta^*\|_\infty$, which often necessitates the incoherence condition [48]. While some studies avoid this by using $\|\hat{\beta} - \beta^*\|_2$ to upper-bound the ℓ_∞ error, this substitution typically requires more stringent minimum signal conditions [44, 19]. In Proposition 2.1 and Theorem 2.2, we depart from these approaches. Instead, by proving the convergence of $\tilde{\beta}^t$ to the oracle estimator $\tilde{\beta}^*$, we establish a sharp ℓ_∞ error bound under only the double sparse RIP condition, thereby eliminating the need for the incoherence-type assumptions required in [11].

Remark 2.3 (Interpretation of the fixed threshold μ). The fixed threshold parameter μ , as defined in (9), is chosen to dominate the statistical error:

- Outside the true support groups, according to Lemma C.3, a threshold of the order $\sqrt{\frac{\sigma^2}{n} \left\{ \frac{\log m}{s_0} + \log(ed/s_0) \right\}}$ ensures that all statistical errors from non-support groups are excluded with high probability.
- Within the true support groups, according to (30) in the proof of Proposition 2.1, a threshold of order $\sqrt{\frac{\sigma^2}{n} \log(sd)}$ ensures that all statistical errors from the support groups are excluded with high probability.

Therefore, by selecting a threshold

$$\mu \asymp \sqrt{\frac{\sigma^2}{n} \max \left\{ \frac{\log m}{s_0} + \log(ed/s_0), \log(sd) \right\}} \asymp \sqrt{\frac{\sigma^2}{n} \left\{ \frac{\log m}{s_0} + \log(sd) \right\}},$$

we effectively filter out statistical errors arising from sub-Gaussian noise. Hence, the second stage iteration could separate the signal from the noise and enforce convergence to the oracle estimator.

Remark 2.4 (Review the DSRIP constant). Under our DSRIP framework, we do not impose the stringent requirement like $\delta < 0.11$ as in [46]; instead, we allow δ to range freely over $(0, 1)$. This generalization theoretically broadens the applicability of our two-stage DSIHT

Algorithm 2. Moreover, our results explicitly quantify the influence of the DSRIP constant δ : As δ increases to 1, the estimation error bounds (8) and (12) inflate, the required signal strength (10) for support recovery grows larger, and the convergence speed in (11) (toward the oracle estimator) becomes slower. These characterizations demonstrate how δ affects both statistical accuracy and computational efficiency. Moreover, by choosing a suitable learning rate in each gradient update step, we show that our DSRIP condition is equivalent to a double sparse Riesz condition, namely, the condition

$$C_L \|u\|_2^2 \leq \frac{1}{n} \|X_S u\|_2^2 \leq C_U \|u\|_2^2, \text{ for every } S \in \mathcal{S}(C_3 s, C_4 s_0) \text{ and } u \in \mathbb{R}^{|S|} \setminus \mathbf{0}_{|S|},$$

where $C_U \geq C_L > 0$ are two arbitrary constants and $C_3, C_4 > 0$ depend on C_U and C_L . This equivalence weakens our original DSRIP assumption and ensures that our procedure can possess rate-optimal results in some random-design settings. Further technical details and proofs are provided in Appendix A.6.

Given that Proposition 2.1 establishes the convergence of estimator $\tilde{\beta}^t$ to the oracle $\tilde{\beta}^*$, it is expected that $\tilde{\beta}_{S^*}^t$ is asymptotically normally distributed as $\tilde{\beta}_{S^*}^*$. For simplicity, we denote by c_ξ the variance of ξ_k for each $k \in [n]$, and denote by B_{S^*} an upper bound on the row-wise ℓ_2 -norm of $X_{S^*} \in \mathbb{R}^{n \times |S^*|}$, i.e., $B_{S^*} := \max_{i \in [n]} \|X_{S^*}^{(i)}\|_2$, where $X_{S^*}^{(i)} \in \mathbb{R}^{|S^*|}$ is the i -th observation of the covariates indexed by S^* .

Theorem 2.3 (Asymptotic Normality). *Assume that all conditions in Proposition 2.1 hold and $B_{S^*}^3 = o_p(\sqrt{n})$. Then, for each fixed $K > 0$ and each matrix $A \in \mathbb{R}^{K \times |S^*|}$, as $n, d, m \rightarrow \infty$, we have*

$$\sqrt{n} A (\tilde{\beta}_{S^*}^t - \beta_{S^*}^*) \xrightarrow{d} N \left(\mathbf{0}, c_\xi \sigma^2 A \left(\frac{1}{n} X_{S^*}^\top X_{S^*} \right)^{-1} A^\top \right).$$

Theorem 2.3 illustrates that the final estimator $\tilde{\beta}^t$ on the true support has an asymptotic distribution identical to that of the oracle estimator $\tilde{\beta}_{S^*}^*$, providing a solid foundation for statistical inference. Notably, Theorem 2.3 remains valid whether ss_0 is fixed or diverging, provided $B_{S^*}^3 = o_p(\sqrt{n})$.

Remark 2.5 (The rate of B_{S^*}). In a fixed-design setting, the rate of B_{S^*} is generally hard to examine. And hence we turn to the random-design setting for a better understanding. Assume that the i -th observation $X^{(i)} \stackrel{d}{=} \Sigma^{1/2} Z^{(i)} \in \mathbb{R}^p$, where $\Sigma \in \mathbb{R}^{p \times p}$ is the population covariance and $Z^{(1)}, \dots, Z^{(n)} \in \mathbb{R}^p$ are i.i.d. centered 1-sub-Gaussian random vectors such that $\mathbf{E}(Z^{(i)} Z^{(i)\top}) = I_p$. In Appendix A.6.3, we prove that $B_{S^*} \lesssim \sqrt{ss_0 + \log n}$ with a probability greater than $1 - 1/n$, and therefore

$$\{B_{S^*}^3 = o_p(\sqrt{n})\} \Leftarrow \{ss_0 = o_p(n^{1/3})\},$$

where the right-hand side gives a more intuitive sufficient condition on the design.

Consequently, by integrating Theorems 2.2 and 2.3, under proper signal conditions, the two-stage DSIHT Algorithm 2 performs as well as if the true support S^* were known in advance, thereby exhibiting the oracle properties (4). These findings enrich the theoretical properties of the double sparse IHT procedure and highlight its superiority over the sparse group Lasso [11].

3. Minimax lower bounds of exact support recovery

In the previous section, we proved that the two-stage DSIHT Algorithm 2 attains exact support recovery at both the group and element levels under the following element-wise and group-wise minimum signal conditions:

$$\begin{aligned} \min_{(i,j) \in S^*} |\beta_{ij}^*| &\geq C \sqrt{\frac{\sigma^2}{n} \left(\frac{\log m}{s_0} + \log(sd) \right)}, \\ \min_{j \in G^*} \|\beta_{G_j}^*\|_2 &\geq C \sqrt{\frac{\sigma^2}{n} (\log m + s_0 \log(sd))}. \end{aligned} \quad (13)$$

In this section, we investigate the necessity of these two conditions. Concretely, we establish minimax lower bounds (under Hamming risk) to show that, if either the element-wise or the group-wise signal condition in (13) fails, then no procedure can simultaneously recover the support set. Thus, the conditions in (13) are necessary for exact recovery in the double sparse model. For simplicity, we assume that $\xi_k \sim N(0, 1)$ independently for every $k \in [n]$.

3.1. How element-wise signal strength affects recovery

To clarify the role of minimum signal strength, we analyze the double sparse space in two parts, first focusing on the element-wise signal strength and then on the group-wise signal strength. Consider a subspace of $\Theta(s, s_0)$ (2) as

$$\Theta_e(s, s_0, a) := \left\{ \beta \in \Theta(s, s_0) \mid \min_{(i,j) \in \text{supp}(\beta)} |\beta_{ij}| \geq a \right\}, \quad (14)$$

where $a > 0$ is a parameter that quantifies the element-wise minimum signal strength. We then construct two subspaces of $\Theta_e(s, s_0, a)$ as

$$\begin{aligned} \Theta_{e,1} &:= \left\{ \beta \in \Theta_e(s, s_0, a) \mid \begin{array}{l} G^*(\beta) = [s], \\ \beta_{ij} = a \text{ for every } (i, j) \in \text{supp}(\beta) \end{array} \right\}, \\ \Theta_{e,2} &:= \left\{ \beta \in \Theta_e(s, s_0, a) \mid \begin{array}{l} \beta_{ij} = a \text{ for every } i \in [s_0], j \in G^*(\beta), \\ \beta_{ij} = 0 \text{ otherwise} \end{array} \right\}. \end{aligned}$$

Specifically, in $\Theta_{e,1}$, the locations of support groups are specified (i.e., the first s groups), while the locations of the support elements are undetermined. In $\Theta_{e,2}$, the locations of the support elements within each support group are specified (i.e., the first s_0 elements), while the locations of the support groups are undetermined. The subspaces $\Theta_{e,1}$ and $\Theta_{e,2}$ provide a concrete understanding of the compositional structure of the double sparse space. The subplots (a) and (b) in Figure 1 provide brief examples of these subspaces.

For ease of display, here we rewrite the element-wise decoder $\eta_{ij}^* = \eta_{ij}(\beta^*) = \mathbf{1}(\beta_{ij}^* \neq 0)$ and the group-wise decoder $(\eta_G^*)_j = (\eta_G)_j(\beta^*) = \mathbf{1}(\beta_{G_j}^* \neq \mathbf{0}_d)$. The Hamming losses

$$\sum_{(i,j) \in [d] \times [m]} |\hat{\eta}_{ij} - \eta_{ij}^*|, \quad \text{and} \quad \sum_{j \in [m]} |(\hat{\eta}_G)_j - (\eta_G^*)_j|$$

are then employed to measure selection errors across different subspaces, where

$$\hat{\eta} = \{\hat{\eta}_{ij}\}_{(i,j) \in [d] \times [m]} \in \{0, 1\}^{d \times m}, \quad \text{and} \quad \hat{\eta}_G = \{(\hat{\eta}_G)_j\}_{j \in [m]} \in \{0, 1\}^m$$

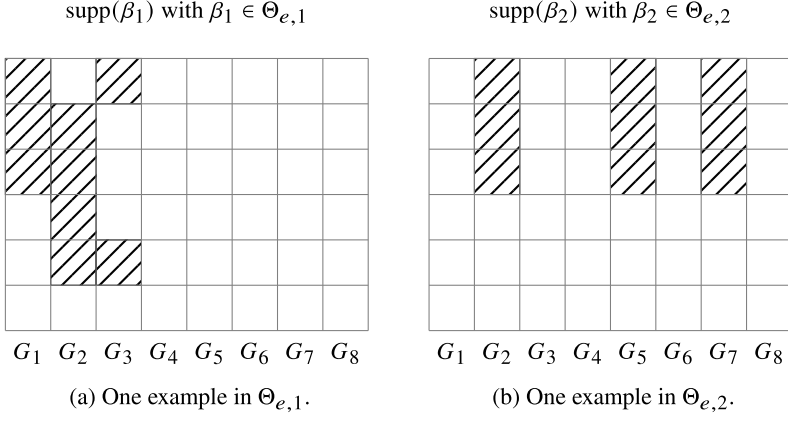


Fig 1. To fully comprehend the subspaces $\Theta_{e,1}, \Theta_{e,2}$, we take two examples β_1, β_2 from them respectively and use the black solid regions to represent their support sets (take $m = 8, d = 6$ and $s = s_0 = 3$). We reshape the group structure as 6×8 matrices with each column representing a group. In $\Theta_{e,1}$ (subfigure (a)), the support groups are the first three groups. In $\Theta_{e,2}$ (subfigure (b)), for each support group, only the first three entries are support entries.

are the element-wise and group-wise selectors aiming to recover the true support pattern.

The following theorem characterizes the minimax lower bounds of selection errors in $\Theta_{e,1}$ and $\Theta_{e,2}$, respectively, demonstrating the complexity of support recovery in the double sparse space $\Theta_e(s, s_0, a)$.

Theorem 3.1 (Minimax lower bounds of selection errors). *Assume that the double sparse model (1) holds with $s < m$ and $s_0 < d$. Then, for every $s'_0 \in (0, s_0)$, we have an element-wise selection lower bound*

$$\begin{aligned}
 & \inf_{\hat{\eta} \in \{0,1\}^{d \times m}} \sup_{\beta^* \in \Theta_e(s, s_0, a)} \mathbf{E}_{Y \sim P_{\beta^*}} \sum_{(i,j) \in [d] \times [m]} |\hat{\eta}_{ij}(Y, X) - \eta_{ij}^*| \\
 & \geq \inf_{\hat{\eta} \in \{0,1\}^{d \times m}} \sup_{\beta^* \in \Theta_{e,1}} \mathbf{E}_{Y \sim P_{\beta^*}} \sum_{(i,j) \in [d] \times [m]} |\hat{\eta}_{ij}(Y, X) - \eta_{ij}^*| \\
 & \geq \frac{ss'_0}{2s_0} \left\{ (d - s_0) \Phi \left(-\frac{a\sqrt{n}}{2\sigma} - \frac{\sigma \log(d/s_0 - 1)}{a\sqrt{n}} \right) \right. \\
 & \quad \left. + s_0 \Phi \left(-\frac{a\sqrt{n}}{2\sigma} + \frac{\sigma \log(d/s_0 - 1)}{a\sqrt{n}} \right) \right\} - 2s(s_0 + s'_0) \exp \left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)} \right). \tag{15}
 \end{aligned}$$

Additionally, for every $s' \in (0, s)$, we have a group-wise selection lower bound

$$\begin{aligned}
 & \inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_e(s, s_0, a)} \mathbf{E}_{Y \sim P_{\beta^*}} \sum_{j \in [m]} |(\hat{\eta}_G)_j(Y, X) - (\eta_G^*)_j| \\
 & \geq \inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_{e,2}} \mathbf{E}_{Y \sim P_{\beta^*}} \sum_{j \in [m]} |(\hat{\eta}_G)_j(Y, X) - (\eta_G^*)_j| \\
 & \geq \frac{s'}{2s} \left\{ (m - s) \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} - \frac{\sigma \log(m/s - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right. \\
 & \quad \left. + s \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} + \frac{\sigma \log(m/s - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right\}
 \end{aligned}$$

$$- 2(s + s') \exp \left(-\frac{3(s - s')^2}{2(s + 2s')} \right). \quad (16)$$

where $\Phi(\cdot)$ is the cumulative distribution function of standard normal distribution, and $\mathbf{E}_{Y \sim P_{\beta^*}}$ represents the expectation for $Y \sim N(X\beta^*, \sigma^2 \mathbf{I}_n)$, as we assume the design matrix X is fixed.

Theorem 3.1 quantifies the influence of the element-wise signal strength a on support recovery in these two subspaces. To avoid confusion with the notation introduced in Section 2, we clarify the distinction in selectors: In Section 2, each element-wise selector $\eta_{ij}(\tilde{\beta}^t) = \mathbf{1}(\tilde{\beta}_{ij}^t \neq 0)$ is defined via the estimator $\tilde{\beta}^t$. By contrast, in Theorem 3.1 we derive lower bounds over arbitrary selectors, not limited to estimator-based selectors, so we employ the more general notation $\hat{\eta}_{ij}(Y, X)$ to emphasize its dependence only on the observed data (Y, X) , rather than on a particular estimator. The proof of Theorem 3.1 is provided in Appendix B. Building on Theorem 3.1, we next give two intuitive minimax lower bounds when a is sufficiently small.

Theorem 3.2 (Necessity of element-wise minimum signal strength). *Assume that the double sparse model (1) holds with $25 \leq s < m/2$, $s_0 < d/2$ and $ss_0 \geq 54$. Assume that the design matrix X satisfies $\text{DSRIP}(s, s_0, \delta)$ with arbitrary $\delta \in (0, 1)$. If the minimum signal strength a satisfies*

$$a^2 \leq \frac{\sigma^2}{10n} \left(\frac{\log(m - s)}{s_0(1 + \delta)} + \log(sd - ss_0) \right), \quad (17)$$

then we have

$$\inf_{\hat{\eta} \in \{0,1\}^{d \times m}} \sup_{\beta^* \in \Theta_e(s, s_0, a)} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{(i,j) \in [d] \times [m]} |\hat{\eta}_{ij} - \eta_{ij}^*| \right\} \geq \frac{(ss_0)^{4/5}}{10}, \quad (18)$$

or

$$\inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_e(s, s_0, a)} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{j \in [m]} |(\hat{\eta}_G)_j - (\eta_G^*)_j| \right\} \geq \frac{s^{7/10}}{20} \quad (19)$$

Theorem 3.2 shows that if (17) holds, then exact support recovery at both element-wise and group-wise levels is impossible: at least one of the two selection tasks must fail. Moreover, by combining (17) with (13), we establish the element-wise minimax separation rate $\sqrt{\frac{\sigma^2}{n} \left(\frac{1}{s_0} \log m + \log(sd) \right)}$ for the double sparse model. By contrast, in the purely element-sparse parameter space $\Theta(ss_0) := \{\beta \in \mathbb{R}^p \mid \|\beta\|_0 \leq ss_0\}$ with $p = dm$, the element-wise minimax separation rate required for exact support recovery is $\sqrt{\frac{\sigma^2}{n} (\log m + \log d)}$ [39, 9], which is larger than the rate above. This gap demonstrates that the double sparse structure facilitates support recovery from the perspective of element-wise signal strength.

Additionally, we stress that our focus is on non-asymptotic minimum signal rates rather than on the asymptotic phase-transition phenomenon of support recovery. Consequently, the numerical constants in Theorem 3.2 are conservative and not optimized for sharpness. The same remark applies to the results in the next subsection.

Remark 3.1 (Interpretation of the element-wise separation rate). We can comprehend the composition of the minimax separation rate (17) from two perspectives:

- Parameter s_0 controls the average number of nonzero entries in each group, and determines the minimum signal strength $\frac{\sigma^2}{n} \frac{\log m}{s_0}$ required for group-wise exact recovery. This term arises from the group selection error in subspace $\Theta_{e,2}$.
- Parameters s and d control the total number of entries in support groups, and determine the minimum signal strength $\frac{\sigma^2}{n} \log(sd)$ required for element-wise exact recovery. This term arises from the element selection error in subspace $\Theta_{e,1}$.

Taking the maximum of these two thresholds yields the separation rate $\frac{\sigma^2}{n} \left\{ \frac{\log m}{s_0} + \log(sd) \right\}$, which provides the lower bound of element-wise signal strength in (17).

3.2. How group-wise signal strength affects recovery

We next analyze how the group-wise signal strength influences the support recovery in the double sparse space $\Theta(s, s_0)$ (2). Define

$$\Theta_g(s, s_0, b) := \left\{ \beta \in \Theta(s, s_0) \mid \min_{j \in G^*(\beta)} \|\beta_{G_j}\|_2 \geq b \right\}, \quad (20)$$

where $b > 0$ is a parameter that quantifies the group-wise minimum signal strength. Similar to Theorems 3.1 and 3.2, we next demonstrate a necessary lower bound of b for the exact support recovery task.

Theorem 3.3 (Necessity of group-wise minimum signal strength). *Assume that the double sparse model (1) holds with $25 \leq s < m/2$, $s_0 < d/2$, $ss_0 \geq 87$, and $s \leq 0.06s_0^{-1/6} \exp(0.15s_0)$. Assume that the design matrix X satisfies DSRIP(s, s_0, δ) with arbitrary $\delta \in (0, 1)$. If b satisfies*

$$b^2 \leq \frac{\sigma^2}{10n} \left(\frac{\log(m-s)}{(1+\delta)} + \frac{s_0}{20} \log(sd - ss_0) \right), \quad (21)$$

then we have

$$\inf_{\hat{\eta} \in \{0,1\}^{d \times m}} \sup_{\beta^* \in \Theta_g(s, s_0, b)} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{(i,j) \in [d] \times [m]} |\hat{\eta}_{ij} - \eta_{ij}^*| \right\} \geq \frac{(ss_0)^{4/5}}{100}, \quad (22)$$

or

$$\inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_g(s, s_0, b)} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{j \in [m]} |(\hat{\eta}_G)_j - (\eta_G^*)_j| \right\} \geq \frac{s^{7/10}}{20}. \quad (23)$$

Theorem 3.3 establishes the group-wise minimax separation rate (measured by the group-wise ℓ_2 norm) as $\sqrt{\frac{\sigma^2}{n} (\log m + s_0 \log(sd))}$. Combining Theorems 3.2 and 3.3, we show that both the element-wise and the group-wise minimum signal conditions, as illustrated in (13), are necessary for exact recovery in the double sparse model: omitting either condition precludes simultaneous support recovery at both element and group levels. These results also confirm that our two-stage DSIHT Algorithm 2 attains minimax rate-optimality in the support recovery problem, thereby demonstrating its theoretical advantage over existing procedures [11, 46].

Remark 3.2 (Minimum sample size for exact recovery). Consider the case that the element-wise and the group-wise minimum signal strengths $\min_{(i,j) \in S^*} |\beta_{ij}^*| = a$ and $\min_{j \in G^*} \|\beta_{G_j}^*\|_2 = g$ are given at firsthand. Then, beyond the optimal sample size $n \gtrsim s s_0 \Delta(s, s_0)$ (required for the DSRIP condition), exact support recovery in the double sparse model further requires

$$n \gtrsim \max \left\{ \frac{\sigma^2}{a^2} \left(\frac{\log m}{s_0} + \log(sd) \right), \frac{\sigma^2}{g^2} (\log m + s_0 \log(sd)) \right\} \\ \asymp \frac{\sigma^2}{\min(a^2, g^2/s_0)} \left(\frac{\log m}{s_0} + \log(sd) \right).$$

Thus, the parameters a^2 and g^2/s_0 jointly influence the sample complexity. The number of support groups s affects the scale of element-wise selection via the $\log(sd)$ term, while the average within-group sparsity s_0 plays a dual role:

- If $s_0 < g^2/a^2$, then s_0 mainly affects the group-selection complexity (through the $(\log m)/s_0$ term).
- If $s_0 \geq g^2/a^2$, then s_0 mainly affects the within-group selection complexity (through the $s_0 \log(sd)$ term).

4. Numerical experiments

This section investigates the finite-sample properties of our proposed algorithm. All simulations are computed using R and executed on a personal laptop with an AMD Ryzen 7 5800H processor operating at 3.20 GHz and 16.00GB of RAM.

Eight estimators are considered in the estimation: The DSIHT method denotes the one-stage Algorithm 1 (proposed by [46]) implemented via the R package ADSIHT, and the Two-stage DSIHT (TS-DSIHT) method is our two-stage procedure (Algorithm 2) that refines the output of DSIHT. The sparse group Lasso (SGLasso) method [35] is fitted using the R package `sparsegl` [24] and its hyperparameters are selected via 5-fold cross-validation. The debiased sparse group Lasso (Debiased-SGLasso) method uses a debiasing technique to refine the SGLasso estimates, following equations (22)-(23) in [11]. The composite MCP (CMCP) method [18] is implemented via the R package `grpreg` [7], tuning by 5-fold cross-validation. Without loss of generality, we also consider the element-wise IHT (IHT-element) and the element-wise Lasso (Lasso-element) as methods under investigation. Finally, we include the Oracle method, i.e., the OLS estimator fitted on the true support, as a baseline for comparison.

We now detail the adaptive implementation of the second stage iteration in Algorithm 2. Based on the double sparse operator $\mathcal{T}_{\mu, s_0}$, it requires determining appropriate element-wise thresholds μ_e and group-wise thresholds μ_g . To select these values adaptively, we consider the set

$$\mathcal{G}_L = \left\{ \left(n^{-\frac{\ell_1-1}{2(L-1)}}, n^{-\frac{\ell_2-1}{2(L-1)}} \right) : 1 \leq \ell_2 < \ell_1 \leq L \right\}, \quad (24)$$

which provides a range of candidate pairs (μ_e, μ_g) based on a geometric series. We then use 5-fold cross-validation to select the optimal pair $(\hat{\mu}_e, \hat{\mu}_g)$ that minimizes prediction error and then refit the second stage iteration on the full data using this choice. This

method is denoted as TS-DSIHT-CV, with $L = 10$. For comparison, we also consider a non-adaptive version, denoted as TS-DSIHT-True, which directly utilizes true parameters (σ, s, s_0) .

4.1. Simulation 1: The influence of signal strength

We first introduce our model setting. Assume that data are generated from the model $y = X\beta^* + \sigma\xi$, where $\sigma = 1$ and $\beta^* \in \mathbb{R}^P$ has $m = 50$ separable groups with equal group size $d = 40$. We randomly draw the design matrix $X \in \mathbb{R}^{n \times P}$ from $N(0, 1)$ independently and standardize by columns to ensure $\|X_{(ij)}\|_2^2 = n$ for each $(i, j) \in [d] \times [m]$, where $n = 300$. We set $s = s_0 = 5$, and

$$\beta_{ij}^* \sim \text{Unif}\left(k\sqrt{\frac{\sigma^2}{n}\left(\frac{\log(m/s)}{s_0} + \log(d/s_0)\right)}, 2k\sqrt{\frac{\sigma^2}{n}\left(\frac{\log(m/s)}{s_0} + \log(d/s_0)\right)}\right) \quad (25)$$

for all $(i, j) \in S^*$ independently, where $k \in [1, 4]$ reflecting the signal strength.

We evaluate the estimation performance through the ℓ_2 error rate $\|\hat{\beta} - \beta^*\|_2 / \|\beta^*\|_2$, and evaluate the support recovery performance using Hamming loss and Matthews Correlation Coefficient [27] at both the group-wise and element-wise levels (shown as Hamming Element, Hamming Group, MCC Element, and MCC Group). We also compare the average computation time of each method.

4.1.1. Estimation and support recovery

Here, we analyze the influence of signal strength on the estimation and support recovery. From Figure 2, it is evident that when the signal strength is less than 0.20, all high-dimensional methods perform poorly, showing substantial deviations from the Oracle estimates. As signal strength increases, both TS-DSIHT-True (using the true (σ, s, s_0)) and the data-driven TS-DSIHT-CV converge toward the Oracle estimator and attain nearly identical estimation accuracy. This indicates that cross-validation can identify suitable threshold pairs $(\hat{\mu}_e, \hat{\mu}_g)$, allowing our output to adaptively achieve the oracle estimation rate when key parameters are unknown in practice. Moreover, both methods recover the true support S^* at both the group-wise and element-wise levels once the signal strength reaches roughly 0.30. Therefore, in the following simulations, we abbreviate the cross-validated procedure TS-DSIHT-CV as TS-DSIHT, referring to the adaptive version of Algorithm 2. The DSIHT method consistently requires stronger signals for similar performance. Although the SGLasso and the CMCP methods show decreasing estimation error rate with stronger signals, both methods remain substantially biased to the Oracle estimation and exhibit poor support recovery performance. The Debiased-SGLasso method removes shrinkage bias and yields asymptotic normality, but it no longer produces sparse estimates, which leads to high ℓ_2 errors and prevents reliable support recovery.

The total runtime of our full two-stage DSIHT method is larger than that of one-stage methods such as SGLasso and CMCP. However, the runtime of our second-stage refinement itself is comparable to those methods. Moreover, the two-stage Debiased-SGLasso directly requires a precision matrix estimation step (see (23) in [11]) that our method avoids, making it substantially slower than ours.

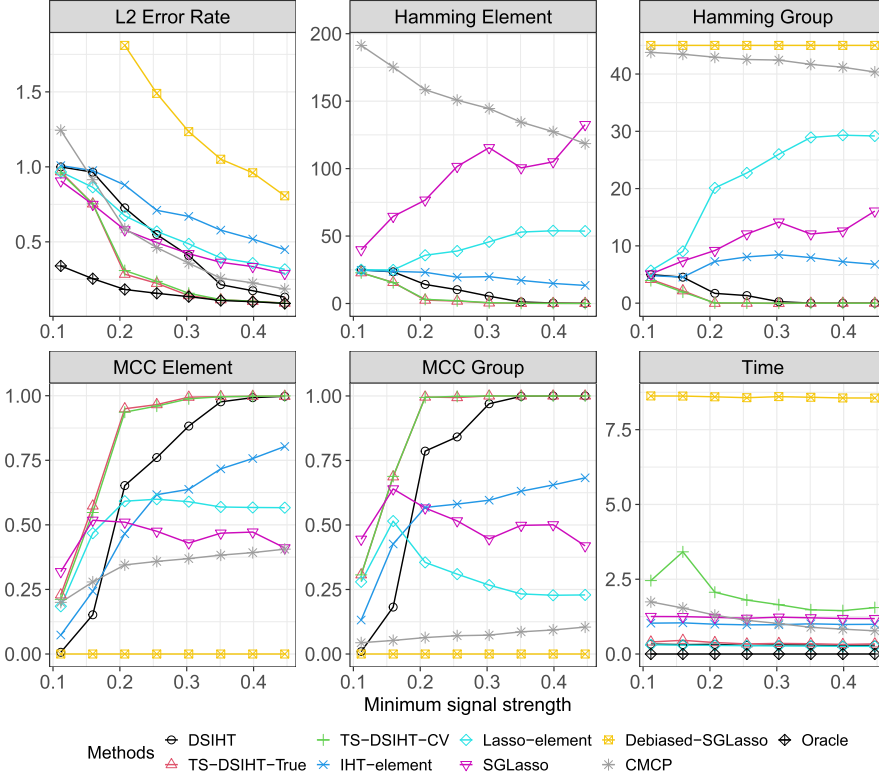


Fig 2. Performance metrics with increasing signal strength. The x-axis represents the minimum signal strength. Each point is averaged from 300 Monte Carlo simulations. The Oracle method (fitted on the true support S^*) has its MCCs always at 1 and Hamming loss always at 0, hence we ignore them. The Debiased-SGLasso produces a desparsified estimator with relatively large ℓ_2 errors; consequently, its extreme values are omitted from the L2 Error Rate plot. Additionally, its element-wise Hamming loss is always at $2000 - 25 = 1975$, so it is omitted from the Hamming Element plot.

4.1.2. Asymptotic property

To assess the asymptotic normality, we set $k = 2.7$ in equation (25) and plot the histograms for $\sqrt{n}(\hat{\beta}_{(1,1)} - \beta_{(1,1)}^*)$ and $\sqrt{n}\sum_{i=1}^5(\hat{\beta}_{(i,3)} - \beta_{(i,3)}^*)$ across sample sizes $n = 300, 600, 900, 1200$, where we denote by $\beta_{i,j}$ the coefficient of the i -th covariate in the j -th group. We also consider using our second-stage iteration in Algorithm 2 (lines 3-6) as a refinement procedure (with 5-fold cross-validation) for the sparse group Lasso estimator, and refer to it as the SGLasso-DSIHT method.

Figure 3 and 4 compare the histograms of the DSIHT, TS-DSIHT, Debiased-SGLasso, and SGLasso-DSIHT methods, superimposed with the Gaussian density curves (red curves). The results show that as the sample size n increases, all four methods exhibit asymptotic normality. However, the TS-DSIHT and SGLasso-DSIHT methods produce distributions that are closer to the normal shape than those of the DSIHT and Debiased-SGLasso methods. This indicates that (i) our proposed second-stage iteration improves asymptotic properties, and (ii) the procedure could be broadly applicable as a refinement step to enhance the statistical behavior of other methods. We provide further evidence for point (ii) in the next experiment.

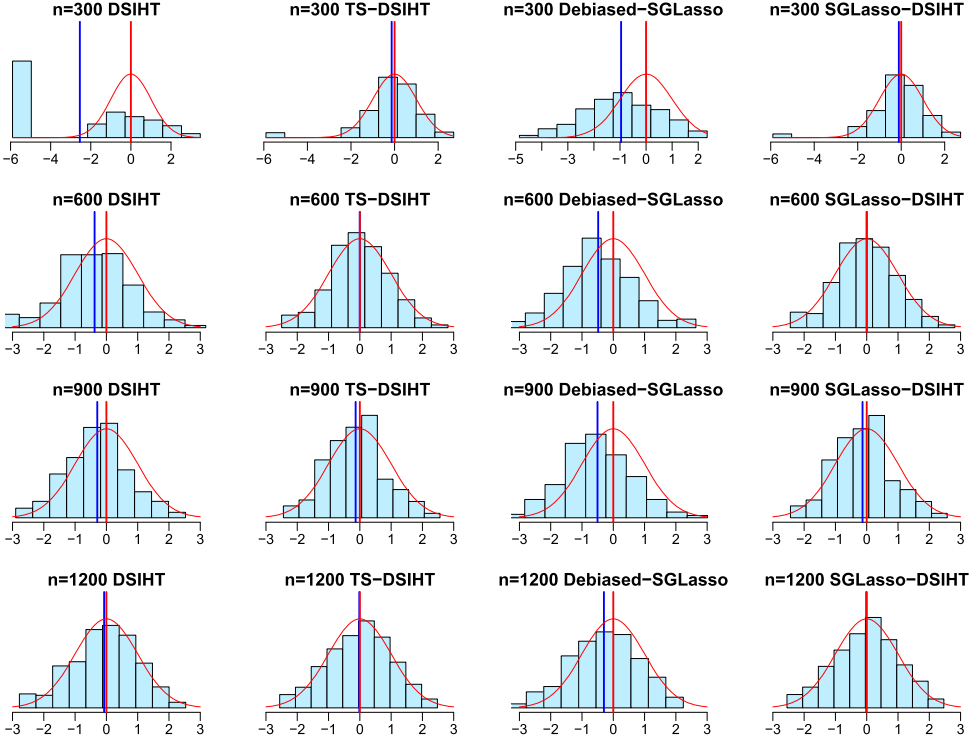


Fig 3. The histograms of $\sqrt{n}(\hat{\beta}_{(1,1)} - \beta_{(1,1)}^*)$ with 300 Monte Carlo simulations conducted for each sample size. Blue vertical lines represent the sample means, and red vertical lines represent the population means.

4.1.3. Universality of the second-stage iteration

Building upon the analysis in Remark 2.1 and the simulation results from Section 4.1.2, we now verify the universality of our second-stage iteration in Algorithm 2. We use the SGLasso and CMCP estimators, respectively, as a warm start for our second-stage algorithm, and then obtain the refined estimators, which are denoted as SGLasso-DSIHT and CMCP-DSIHT. We adopt the same parameter settings as in Section 4.1.1, and the simulation results are presented in Figure 5. The SGLasso-DSIHT and CMCP-DSIHT estimators demonstrate substantially better performance in both estimation and support recovery compared to their initial estimators: As the signal strength increases, these refined estimators, along with TS-DSIHT, all achieve the oracle estimation rate and exact support recovery. This indicates that our proposed second-stage iteration is effective and has a general applicability as a refinement method to debias or enhance other estimators.

4.2. Simulation 2: The influence of sparsity levels

This subsection analyzes the influence of sparsity level s and s_0 in the estimation and support recovery. We assume $n = 500$ and $ss_0 = 48$, and the remaining parameters $d = 40$, $m = 50$, $\sigma = 1$ are identical to those in Section 4.1. We consider the double sparse spaces under the following three cases, respectively:

- **Case A.** $s_0 = 4$, $s = 12$.

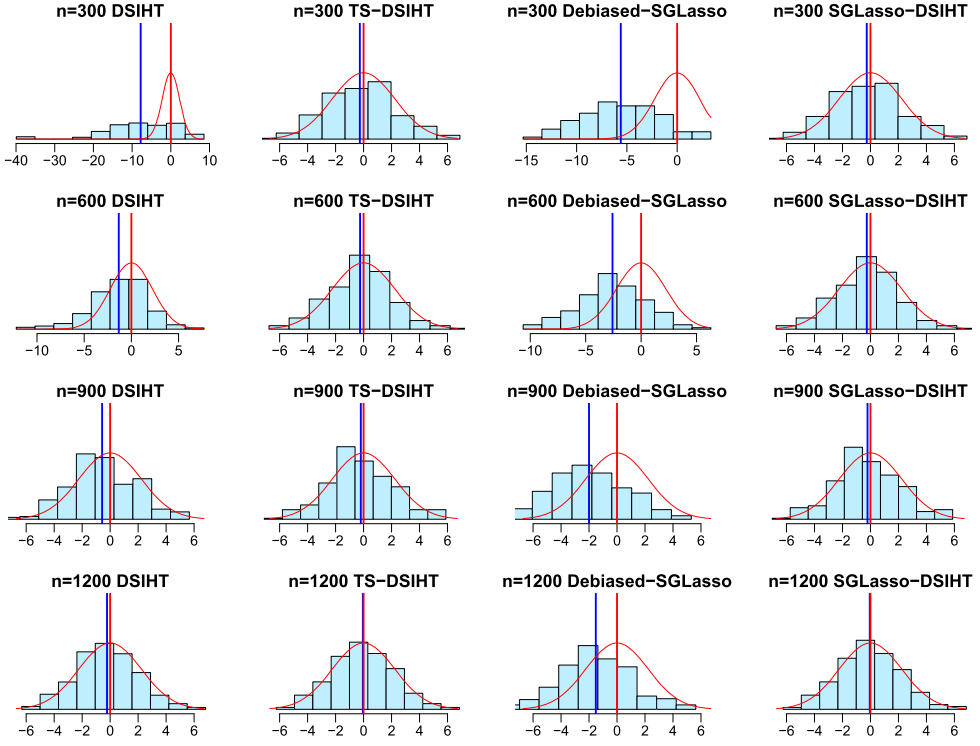


Fig 4. The histograms of $\sqrt{n} \sum_{i=1}^5 (\hat{\beta}_{(i,3)} - \beta_{(i,3)}^*)$ with 300 Monte Carlo simulations conducted for each sample size. Blue vertical lines represent the sample means, and red vertical lines represent the population means.

- **Case B.** $s_0 = 8$, $s = 6$.
- **Case C.** $s_0 = 12$, $s = 4$.

To mitigate confounding effects, we assume equal signal strength across support sets in both cases, i.e., $\beta_{ij}^* = a$ for all (i, j) in the support set of each case. We now analyze the performance of TS-DSIHT (Algorithm 2) under these three cases (TS-DSIHT A, TS-DSIHT B, and TS-DSIHT C), and assess their deviation from the corresponding Oracle estimators (Oracle A, Oracle B, and Oracle C). The ℓ_2 error rate $\|\hat{\beta} - \beta^*\|_2 / \|\beta^*\|_2$, element-wise and group-wise Hamming loss, and Matthews Correlation Coefficients are used to measure the effectiveness.

With the fixed $ss_0 = 48$, Figure 6 shows that a larger s_0 (Case C) leads to a weaker elemental signal strength required for the element-wise recovery (see subplots “Hamming Element” and “MCC Element”), consistent with the element-wise minimum signal condition in Proposition 2.1, i.e., of the signal rate $\frac{\sigma}{\sqrt{n}} \sqrt{\frac{\log m}{s_0} + \log(sd)}$. However, a larger s_0 also results in a stronger group signal strength required for the group recovery, consistent with the group-wise minimum signal condition in Proposition 2.1, i.e., of the rate $\frac{\sigma}{\sqrt{n}} \sqrt{\log m + s_0 \log(sd)}$. Therefore, Case A is favored in support group recovery (see subplots “Hamming Group” and “MCC Group”). Hence, as the intra-group sparsity s_0 increases, achieving equivalent support recovery outcomes necessitates a more stringent requirement on group strength, while the requirement on element-wise strength becomes more relaxed.

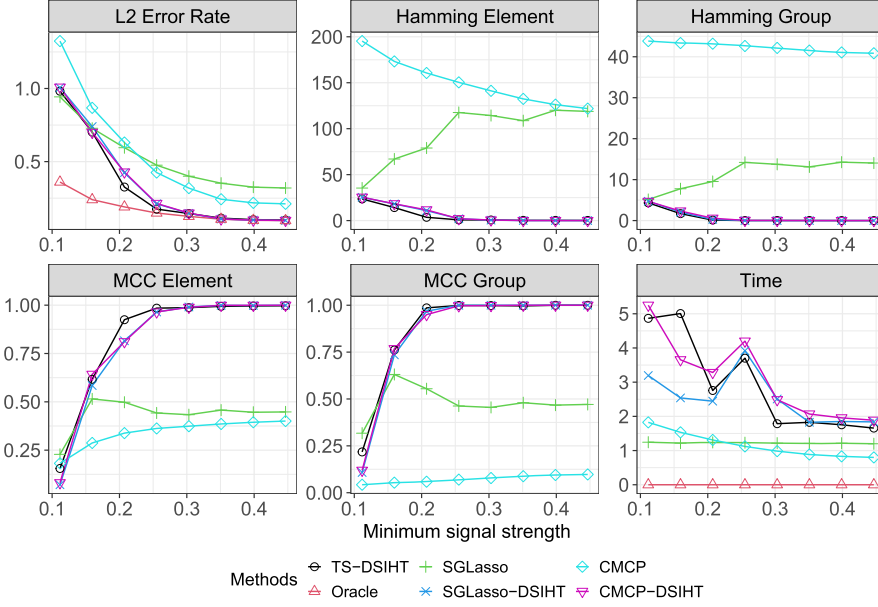


Fig 5. Refinement effect of the second-stage DSIHT with increasing signal strength. Each point is averaged from 300 Monte Carlo simulations. In any method that incorporates the second-stage DSIHT iteration, this specific part relies on the grid points in (24) and employs 5-fold cross-validation for data-driven estimation.

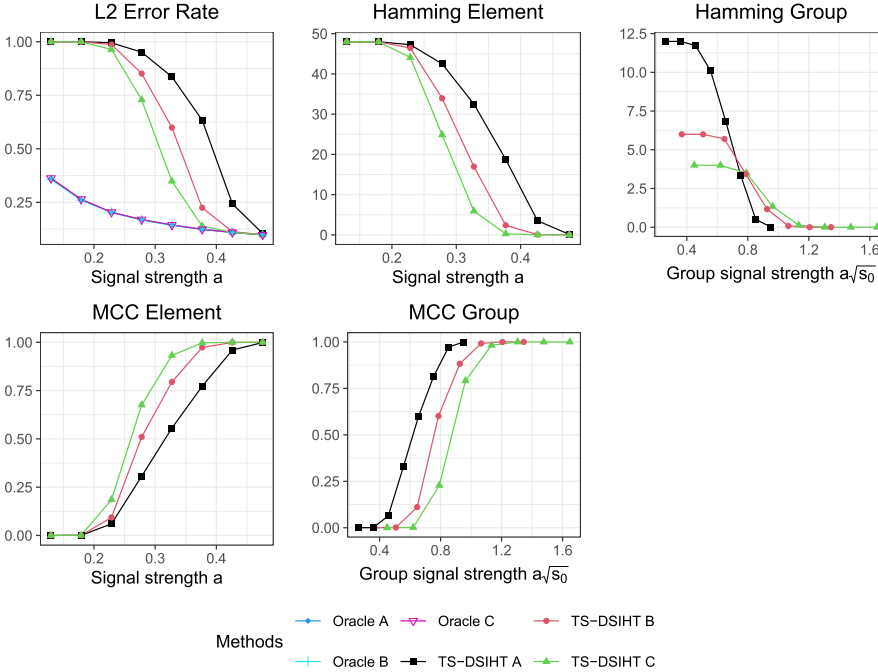


Fig 6. Performance metrics with different sparsity levels. The x-axis in the first two subplots represents the element signal strength a , and in the third subplot, it represents the group signal strength $a\sqrt{s_0}$. Each point is averaged from 300 Monte Carlo simulations.

4.3. Real data analysis

We finally evaluate these methods on a supermarket sales dataset [40] containing $n = 464$ daily observations and 6,398 product-level sales volume. The response is the daily customer count. We standardize all variables and keep the 300 predictors with the largest absolute Pearson correlation with the response. Each selected predictor is then expanded into an 8-term B-spline basis, yielding a double sparse design with $m = 300$ and $d = 8$ that can capture flexible, potentially nonlinear effects of product sales.

For performance assessment, we performed 100 random 80/20 train/test splits (371 observations for training and 93 for testing in each split). Each fitted model is measured on the test set using the mean squared prediction error (MSPE). Table 2 reports the mean and standard deviation (over 100 random splits) of MSPE, the number of selected groups, and the number of selected elements. TS-DSIHT achieves the lowest average MSPE and selects a substantially smaller model than most of the competing methods. These results underscore the practical advantages of our method: better predictive accuracy together with a relatively interpretable model.

Table 2

Results of each method based on 100 random partitions. The standard deviation is shown in parentheses

Method	MSPE	Number of selected groups	Number of selected elements
TS-DSIHT	0.468 (0.105)	3.860 (3.315)	21.240 (15.588)
DSIHT	0.503 (0.121)	2.290 (1.066)	8.510 (4.596)
SGLasso	0.483 (0.100)	9.930 (2.6789)	42.530 (11.788)
Debiased-SGLasso	0.914 (0.183)	300 (0)	2400 (0)
CMCP	0.507 (0.121)	50.920 (5.134)	63.020 (6.734)
IHT-element	0.782 (0.182)	223.980 (124.509)	225.240 (125.172)
Lasso-element	0.478 (0.093)	14.270 (4.194)	17.840 (5.773)

Note: Debiased-SGLasso yields a dense solution and therefore selects all 300 groups and 2,400 variables in every split.

5. Conclusion and discussion

In this paper, we propose the specific minimum signal conditions that serve as both the sufficient and necessary conditions for exact recovery in the double sparse model. From the perspective of sufficiency, we demonstrate that a two-stage DSIHT algorithm achieves oracle properties under these minimum signal conditions, ensuring exact support recovery and asymptotic normality. From the perspective of necessity, we show through minimax lower bounds that no algorithm can achieve exact recovery if these signal conditions are violated. Our work fills a critical gap in the minimax optimality theory on support recovery of the double sparse model, and establishes the minimax rate-optimality of the double sparse IHT procedure for support recovery. However, some constants in this paper are not exact, meaning that practical implementations of the algorithm do not need to adhere to them strictly.

Additionally, Proposition 2.1 and Theorem 2.3 reveal an advantage of the IHT-type estimator over convex penalized estimators like the Lasso: The IHT-type algorithm permits precise control over estimation error in each iteration step, facilitating an in-depth analysis of the relationship between the signal condition and support recovery performance. We believe

this intuitive non-convex approach could be effective in a broader range of models, including multi-attribute graphical models, sparse plus low-rank matrix regression, and other general problems under double sparse structure. We leave these meaningful problems for future work.

Appendix A: Proof of the oracle properties

This appendix focuses on the proofs of the upper bound. Firstly, we introduce some abbreviations.

A.1. Some abbreviations and preliminaries

Table 3 introduces some useful abbreviations in the following proof. For ease of display, recall that we use the double index to locate the entry in β^* , i.e., use β_{ij}^* to represent the i -th signal in the j -th group G_j .

Table 3
Symbols and their meanings

Symbol	Meaning
$S(as, bs_0)$	The space consisting of all the support sets of (as, bs_0) -sparse vectors.
$S^* = \{(i, j) : \beta_{ij}^* \neq 0\}$	The true support set of β^* .
$G^* = \{j : \beta_{G_j}^* \neq \mathbf{0}_{ G_j }\}$	The group index set of the true support groups.
$S_{G^*} = \bigcup_{j \in G^*} G_j$	The index set of all support groups.
$s_j = \ \beta_{G_j}^*\ _0 = S^* \cap G_j $	The sparsity level, i.e., the support number in group G_j .
$\tilde{S}^t$	The support set of the estimation $\tilde{\beta}^t$ (in the t -th iteration) by Algorithm 2.
$\tilde{S}_{OG}^t = (S_{G^*})^c \cap \tilde{S}^t$	The discovered set in the falsely discovered groups in the t -th iteration (by Algorithm 2).
$\tilde{S}_{IG}^t = S_{G^*} \cap (S^*)^c \cap \tilde{S}^t$	The falsely discovered set in the true support groups in the t -th iteration (by Algorithm 2).
$\tilde{G}_{OG}^t$	The group index set of the falsely discovered group in the t -th iteration (by Algorithm 2).
$\Delta(s, s_0) := \frac{1}{s_0} \log(em/s) + \log(ed/s_0)$	A useful abbreviation associated with ordinary minimax rate.

To simplify the notations, we use the double index (i, j) to locate the i -th variable in the j -th group G_j . In the beginning, we introduce an essential definition $\tilde{H}^{t+1} := \tilde{\beta}^t + \frac{1}{n} X^\top (Y - X\tilde{\beta}^t) \in \mathbb{R}^P$. It can be decomposed as follows:

$$\begin{aligned} \tilde{H}^{t+1} &= \tilde{\beta}^t + \frac{1}{n} X^\top (Y - X\tilde{\beta}^t) \\ &= \tilde{\beta}^t + \frac{1}{n} X^\top (X\tilde{\beta}^* + \sigma\tilde{\xi} - X\tilde{\beta}^t) = \tilde{\beta}^* + \Phi(\tilde{\beta}^* - \tilde{\beta}^t) + \tilde{\Xi}, \end{aligned} \quad (26)$$

where $\Phi = \frac{1}{n} X^\top X - \mathbf{I}_p \in \mathbb{R}^{p \times p}$. $\tilde{\beta}^*$ is the oracle estimator, that is, $\tilde{\beta}_{S^*}^* := (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top Y \in \mathbb{R}^{|S^*|}$, and $\tilde{\beta}_{(S^*)^c}^* = \mathbf{0}$. And $\sigma\tilde{\xi} := Y - X\tilde{\beta}^*$. Define

$$\tilde{\Xi} := \frac{\sigma}{n} X^\top \left(\mathbf{I}_n - X_{S^*} (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top \right) \xi \in \mathbb{R}^P.$$

By sub-Gaussian property of ξ , we conclude that $\tilde{\Xi}_{ij} = 0$ holds for all $(i, j) \in S^*$, and $\mathbf{E} \exp(\lambda \tilde{\Xi}_{ij}) \leq \exp(\lambda^2 \sigma^2 / (2n))$ holds for all $\lambda \in \mathbb{R}$ and $(i, j) \in (S^*)^c$. Besides, by $\tilde{\beta}_{S^*}^* - \beta_{S^*}^* = \sigma(X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top \xi$, we have $\mathbf{E} \exp(\lambda(\tilde{\beta}_{ij}^* - \beta_{ij}^*)) \leq \exp\left(\frac{\lambda^2 \sigma^2}{2(1-\delta)n}\right)$ holds for all $\lambda \in \mathbb{R}$ and $(i, j) \in S^*$.

For ease of display, we denote $\Phi_{(k,j)} \in \mathbb{R}^{1 \times p}$ as the row of Φ corresponding to the k -th variable in the j -th group, i.e., $\Phi_{(k,j)}$ is the $((j-1)d+k)$ -th row vector of $\Phi \in \mathbb{R}^{p \times p}$. Plus,

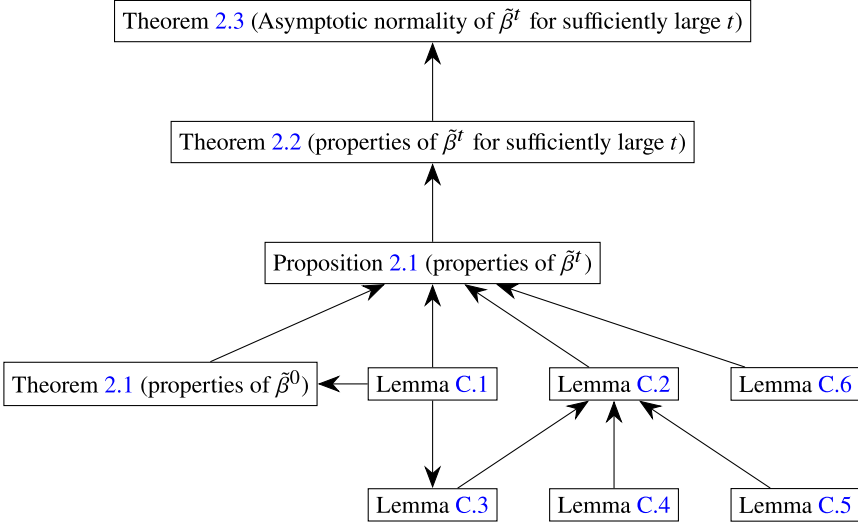


Fig 7. A proof flowchart for conclusions in Section 2. The directed edges mean that the tails of the edges are used in the heads of the edges.

we show a technique that will be frequently used in the following proof. Recall $A = A(\kappa, \delta) := \frac{8\delta^2}{(\kappa - \delta)^2}$. For every $\tilde{\beta}^* - \tilde{\beta}^t$ and S satisfying $S \cup \text{supp}(\tilde{\beta}^* - \tilde{\beta}^t) \in \mathcal{S}\left((1 + 2A)s, \frac{1+4A}{1+2A}s_0\right)$, based on $\text{DSRIP}\left((1 + 2A)s, \frac{1+4A}{1+2A}s_0, \delta\right)$ condition (recall Definition 1.1), we have

$$\begin{aligned}
 \sum_{(k,j) \in S} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 &\leq \sum_{(k,j) \in S'} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 \\
 &= (\tilde{\beta}^* - \tilde{\beta}^t)_{S'}^\top \Phi_{S'S'}^\top \Phi_{S'S'} (\tilde{\beta}^* - \tilde{\beta}^t)_{S'} \\
 &\leq \|\Phi_{S'S'}\|_2^2 \cdot \|\tilde{\beta}^* - \tilde{\beta}^t\|_2^2 \\
 &\leq \delta^2 \|\tilde{\beta}^* - \tilde{\beta}^t\|_2^2,
 \end{aligned} \tag{27}$$

where $S' = S \cup \text{supp}(\tilde{\beta}^* - \tilde{\beta}^t)$.

In the following subsections, we first prove Proposition 2.1, Theorem 2.2, and Theorem 2.3, and then give a brief proof of Theorem 2.1 (because it can be seen as a simpler version of Proposition 2.1). The relationship between Proposition 2.1, Theorem 2.1, and some lemmas in Figure 7 is illustrated in Figure 7. The final subsection A.6 provides additional discussion on the random setting.

A.2. Proof of Proposition 2.1

Preliminary Define

$$\mu := \max \left\{ \frac{\kappa C_\lambda}{\delta}, \sqrt{40 + \frac{120\delta^2}{(1 - \delta)^2}} \right\} \cdot \sqrt{\frac{\sigma^2}{n} \left\{ \frac{\log(em)}{s_0} + \log(esd) \right\}}, \tag{28}$$

where C_λ and $\lambda_{(\infty)}$ are related to the parameter setting in Theorem 2.1. The definition of μ will be used in (34), Lemma C.2, and Lemma C.3.

First, we show the probability inequality used in this proof as follows.

$$P \left\{ \begin{array}{l} \sup_{S \in \mathcal{S}(1, s_0)} \|\tilde{\Xi}_S\|_2^2 \leq \frac{10\sigma^2 s_0 \Delta(1, s_0)}{n}, \\ \|\tilde{\beta}^* - \beta^*\|_2^2 \leq \frac{5\sigma^2 s s_0 \Delta(s, s_0)}{n(1-\delta)}, \\ \max_{(i,j) \in S^*} |\tilde{\beta}_{ij}^* - \beta_{ij}^*| \leq \frac{\mu}{3}, \\ \max_{(i,j) \in S_{G^*}} |\tilde{\Xi}_{ij}| \leq \underbrace{\sqrt{\frac{10\sigma^2}{n} [\Delta(1, s_0) + \log(ss_0)]}}_{=:\mu'} \end{array} \right\} \geq 1 - O\left(e^{-\frac{1}{3}[\Delta(1, s_0) + \log(ss_0)]}\right), \quad (29)$$

where the first inequality follows from Lemma C.1, with taking $s' = 1, s'_0 = s_0$, the second inequality follows from a similar result as in (67), with $\|X_{S^*}^\top X_{S^*}\|_2 \leq \frac{1}{n(1-\delta)}$ and taking $t = ss_0 \Delta(s, s_0)$. The third and fourth inequalities follow from the sub-Gaussian property, for example, since

$$\begin{aligned} \frac{\mu}{3} &\geq \frac{1}{3} \sqrt{4 + \frac{12\delta^2}{(1-\delta)^2}} \cdot \sqrt{\frac{10\sigma^2 [\Delta(1, s_0) + \log(ss_0)]}{n}} \\ &\geq \frac{\sqrt{30}\sigma}{3(1-\delta)} \sqrt{\frac{[\Delta(1, s_0) + \log(ss_0)]}{n}}, \end{aligned}$$

we have

$$\begin{aligned} P \left(\max_{(i,j) \in S^*} |\tilde{\beta}_{ij}^* - \beta_{ij}^*| \geq \frac{\mu}{3} \right) &= P \left(\bigcup_{(i,j) \in S^*} \left\{ |\tilde{\beta}_{ij}^* - \beta_{ij}^*| \geq \frac{\mu}{3} \right\} \right) \\ &\leq 2s s_0 \cdot \exp \left(-\frac{5 [\Delta(1, s_0) + \log(ss_0)]}{3(1-\delta)} \right) \\ &\leq 2 \exp \left\{ -\frac{1}{3} [\Delta(1, s_0) + \log(ss_0)] \right\}. \end{aligned}$$

Additionally,

$$P \left(\max_{(i,j) \in S_{G^*}} |\tilde{\Xi}_{ij}| \geq \mu' \right) \leq 2sd \cdot \exp \left(-\frac{n\mu'^2}{2\sigma^2} \right) = 2 \exp \left(-\frac{n\mu'^2}{2\sigma^2} + \log(sd) \right). \quad (30)$$

Mathematical induction With $A = \frac{8\delta^2}{(\kappa-\delta)^2}$, we next prove the following results hold for all $t \geq 0$:

$$\begin{aligned} \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 &\leq \left[\sqrt{5/6} + \left(1 - \sqrt{5/6}\right) \delta \right]^t \|\tilde{\beta}^0 - \tilde{\beta}^*\|_2, \\ \tilde{S}_{OG}^t &:= (S_{G^*})^c \cap \tilde{S}^t \in \mathcal{S}(As, s_0), \\ \tilde{S}_{IG}^t &:= S_{G^*} \cap \tilde{S}^t \cap (S^*)^c \in \mathcal{S}(s, As_0), \end{aligned} \quad (31)$$

Under the event mentioned in (29), we prove (31) by using mathematical induction. We first check (31) in $t = 0$: It is straightforward that the first results hold, and by Theorem 2.1, the

initial estimator $\tilde{\beta}^0$ satisfies $\tilde{S}_{OG}^t \in \mathcal{S}(As, s_0)$ and $\tilde{S}_{IG}^t \in \mathcal{S}(s, As_0)$ with a probability greater than $1 - O\left(e^{-\frac{1}{3}[\Delta(1, s_0) + \log(s s_0)]}\right)$.

Then, we assume that (31) holds in the t -th iteration (for some $t \geq 0$), and we need to prove that (31) still holds in the $(t+1)$ -th iteration. Next two steps prove $\tilde{S}_{OG}^{t+1} \in \mathcal{S}(As, s_0)$ and $\tilde{S}_{IG}^{t+1} \in \mathcal{S}(s, As_0)$ in $(t+1)$ -th iteration. The way is to prove by contradiction.

Step 1 (Control the shape of $\tilde{S}_{OG}^{t+1}$). By using contradiction, at first we assume the opposite, that is, $\tilde{S}_{OG}^{t+1}$ contains more than As groups (i.e., these groups are not true support groups, but their coefficients are not estimated as zero in the $(t+1)$ -th iteration). Then, we can select arbitrary As falsely discovered groups and construct an index set $S'_{OG} \in \mathcal{S}(As, s_0)$. The details of the construction process are described as follows:

For any falsely discovered group G_j (i.e., $j \notin G^*$), if $\|\tilde{\beta}_{G_j}^{t+1}\|_0 \geq s_0$, then choose arbitrarily s_0 non-zero entries from $\tilde{\beta}_{G_j}^{t+1}$ into S'_{OG} ; if $\|\tilde{\beta}_{G_j}^{t+1}\|_0 < s_0$, then choose all non-zero entries from $\tilde{\beta}_{G_j}^{t+1}$ into S'_{OG} . Repeat this operation As times for any As falsely discovered groups, and we obtain a (As, s_0) -sparse set S'_{OG} , i.e., $S'_{OG} \in \mathcal{S}(As, s_0)$.

Then, based on the element-wise thresholding operator $\mathcal{T}_\mu^{(1)}$ and the group-wise thresholding operator $\mathcal{T}_{\mu, s_0}^{(2)}$, for any group G_j selected by S'_{OG} , it holds that

$$\|\tilde{\beta}_{G_j \cap S'_{OG}}^{t+1}\|_2^2 = \left\| \mathcal{T}_{\mu, s_0} \left(\tilde{H}^{t+1} \right)_{G_j \cap S'_{OG}} \right\|_2^2 \geq s_0 \mu^2.$$

Based on the decomposition $\tilde{H}_{ij}^{t+1} = \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle + \Xi_{ij}$ (holds for all $(i, j) \in (S^*)^c$) and triangle inequality, we derive that

$$\begin{aligned} \sqrt{Ass_0} \mu &\leq \sqrt{\sum_{(i,j) \in S'_{OG}} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2} + \sqrt{\sum_{(i,j) \in S'_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\}} \\ &\stackrel{(i)}{\leq} \delta \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 + \frac{1-\delta}{\sqrt{3}} \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 \\ &< \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 \\ &\stackrel{(ii)}{\leq} \|\tilde{\beta}^0 - \beta^*\|_2 + \|\beta^* - \tilde{\beta}^*\|_2 \\ &\stackrel{(iii)}{\leq} \left(1 - \frac{\sqrt{10}}{C_\lambda}\right) \frac{2\sqrt{2}\kappa}{\kappa - \delta} \cdot \sqrt{ss_0} \lambda_{(\infty)} + \frac{\kappa}{\kappa - \delta} \sqrt{5ss_0} \frac{\lambda_{(\infty)}}{C_\lambda} \\ &< \sqrt{Ass_0} \mu, \end{aligned} \tag{32}$$

where inequality (i) follows from (27) and Lemma C.2, inequality (ii) follows from the first result in (31) in the t -th iteration (held by the assumption of induction). Inequality (iii) follows from Theorem 2.1 and the second inequality in (29), and the final inequality follows the definition of μ , as shown in (28). Therefore, (32) leads to an absurdity, demonstrating that only fewer than As groups can be falsely discovered in the $(t+1)$ -th iteration.

Next, we prove that fewer than Ass_0 elements will be falsely discovered outside true groups G^* . If not so, we can construct set $S''_{OG} \in \mathcal{S}(As, s_0)$, which satisfies that $S''_{OG} \subseteq \tilde{S}_{OG}^{t+1}$, $|S''_{OG}| = Ass_0$ and each $\tilde{\beta}_{ij}^{t+1}$ indexed from S''_{OG} is falsely discovered. The construction is similar to S'_{OG} : For any falsely discovered group G_j , if $\|\tilde{\beta}_{G_j}^{t+1}\|_0 < s_0$, then choose all non-zero entries from $\tilde{\beta}_{G_j}^{t+1}$ into S''_{OG} . If $\|\tilde{\beta}_{G_j}^{t+1}\|_0 \geq s_0$, then choose at least s_0 of its nonzero entries

in the set S''_{OG} —crucially, we carefully calibrate the exact number of selections across all such groups so that the total cardinality $|S''_{OG}| = Ass_0$. Therefore, based on the element-wise thresholding operator and Lemma C.2, we have

$$\begin{aligned} \sqrt{Ass_0}\mu &\leq \sqrt{\sum_{(i,j) \in S''_{OG}} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2} + \sqrt{\sum_{(i,j) \in S''_{OG}} \Xi_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu,s_0}(\tilde{H}^{t+1})_{ij} \neq 0\}} \\ &\leq \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 < \sqrt{Ass_0}\mu. \end{aligned} \quad (33)$$

Then, we similarly find an absurdity as (32) again. Therefore, we prove that $\tilde{S}_{OG}^{t+1} = S_{G^*}^c \cap \tilde{S}^{t+1} \in \mathcal{S}(As, s_0)$.

Step 2 (Control the shape of $\tilde{S}_{IG}^{t+1}$).

By using contradiction, at first we assume the opposite, that is, more than Ass_0 entries in S_{G^*} are falsely discovered. Then we can construct an index set $S'_{IG} \subseteq \tilde{S}_{IG}^{t+1}$ satisfying $|S'_{IG}| = Ass_0$ and $S'_{IG} \in \mathcal{S}(s, As_0)$, with $|\mathcal{T}_{\mu,s_0}(\tilde{H}^{t+1})_{ij}| \geq \mu$ for all $(i, j) \in S'_{IG}$. Therefore, based on the element-wise thresholding operator, we obtain

$$\begin{aligned} \sqrt{Ass_0}\mu &\leq \sqrt{\sum_{(i,j) \in S'_{IG}} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2} + \sqrt{\sum_{(i,j) \in S'_{IG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij} + \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu\}} \\ &\leq \delta \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 + \sqrt{\sum_{(i,j) \in S'_{IG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| \geq \mu'\}} \\ &\quad + \sqrt{\sum_{(i,j) \in S'_{IG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| < \mu', |\tilde{\Xi}_{ij} + \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu\}} \\ &\stackrel{(i)}{\leq} \delta \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 + \sqrt{\sum_{(i,j) \in S'_{IG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| < \mu' \leq \frac{1-\delta}{\delta\sqrt{6}} |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle|\}} \\ &\leq \|\tilde{\beta}^t - \tilde{\beta}^*\|_2, \end{aligned} \quad (34)$$

where recall $\mu' = \sqrt{\frac{10\sigma^2}{n} [\Delta(1, s_0) + \log(ss_0)]}$ and $\mu \geq \sqrt{4 + \frac{12\delta^2}{(1-\delta)^2}} \cdot \mu'$, and inequality (i) follows from the forth inequality in (29). Then, by the last three lines in (32), we get a contradiction again, demonstrating $\tilde{S}_{IG}^{t+1} = S_{G^*} \cap \tilde{S}^{t+1} \cap (S^*)^c \in \mathcal{S}(s, As_0)$.

Step 3 (ℓ_2 error inequality of $\tilde{\beta}^{t+1}$). Now we prove that the ℓ_2 error bound in the $(t+1)$ -th iteration. Note that

$$\begin{aligned} &\tilde{\beta}_{ij}^{t+1} - \tilde{\beta}_{ij}^* \\ &= - \underbrace{(\tilde{\beta}_{ij}^* + \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle + \tilde{\Xi}_{ij}) \cdot \mathbf{1}\left((i, j) \notin \tilde{S}^{t+1}\right)}_{\tilde{H}_{ij}^{t+1}} + \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle + \tilde{\Xi}_{ij}, \end{aligned} \quad (35)$$

where recall $\tilde{\Xi}_{ij} = 0$ for all $(i, j) \in S^*$. We then have

$$\begin{aligned}
& \|\tilde{\beta}^{t+1} - \tilde{\beta}^*\|_2 \\
& \leq \left[\sum_{(i,j) \in S^*} \left\{ \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle - \tilde{H}_{ij}^{t+1} \cdot \mathbf{1}((i,j) \notin \tilde{S}^{t+1}) \right\}^2 \right. \\
& \quad \left. + \sum_{(i,j) \in \tilde{S}^{t+1} \setminus S^*} \left\{ \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle + \tilde{\Xi}_{ij} \right\}^2 \right]^{1/2} \\
& \stackrel{(i)}{\leq} \sqrt{\sum_{(i,j) \in \tilde{S}^{t+1} \cup S^*} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2} \\
& \quad + \left[\sum_{(i,j) \in \tilde{S}_{OG}^{t+1}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} + \sum_{(i,j) \in \tilde{S}_{IG}^{t+1}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \right. \\
& \quad \left. + \sum_{(i,j) \in S^*} \left\{ \tilde{H}_{ij}^{t+1} \right\}^2 \mathbf{1}((i,j) \notin \tilde{S}^{t+1}) \right]^{1/2} \\
& \stackrel{(ii)}{\leq} \left(1 + \sqrt{\frac{5}{6}} \frac{1-\delta}{\delta} \right) \delta \|\tilde{\beta}^t - \tilde{\beta}^*\|_2 \\
& \leq \left[\sqrt{5/6} + (1 - \sqrt{5/6}) \delta \right]^{t+1} \|\tilde{\beta}^0 - \tilde{\beta}^*\|_2,
\end{aligned} \tag{36}$$

where inequality (i) follows from the triangle inequality, and inequality (ii) follows from Lemma C.2, (34), and Lemma C.6. The last inequality follows from the first result in (31) in the t -th iteration, which holds by the assumption of induction.

Therefore, we prove that all three results in (31) still hold in the $(t+1)$ -th iteration. This completes the proof of Proposition 2.1.

A.3. Proof of Theorem 2.2

We follow the results derived in Proposition 2.1. By the sample size assumption and inequalities (ii), (iii) in (32), we have $\|\tilde{\beta}^0 - \tilde{\beta}^*\|_2 < \sqrt{A s s_0} \mu \leq C$ for a sufficiently large absolute constant $C > 0$, therefore

$$\|\tilde{\beta}^t - \tilde{\beta}^*\|_2 \leq \left[\sqrt{5/6} + (1 - \sqrt{5/6}) \delta \right]^t \cdot C\sigma, \quad \text{for every } t \geq 0. \tag{37}$$

Then, let $t > C_\delta \log n$, and we conclude

$$\|\tilde{\beta}^t - \tilde{\beta}^*\|_2 < \frac{\sigma}{n}, \tag{38}$$

which leads to

$$\|\tilde{\beta}^t - \beta^*\|_\infty \leq \|\tilde{\beta}^t - \tilde{\beta}^*\|_\infty + \|\tilde{\beta}^* - \beta^*\|_\infty < \mu/2, \tag{39}$$

where the last inequality follows from the third inequality in (29) and $\mu > \frac{6\sigma}{\sqrt{n}}$.

Then, we conclude that:

1. For every $(i, j) \in (S^*)^c$, by (39), we have $|\tilde{\beta}_{ij}^t| < \mu/2 < \mu$, which proves that we can not falsely discover any entry by using thresholding parameter μ (just recall the element-wise operator $\mathcal{T}_\mu^{(1)}$), therefore $\tilde{\beta}_{ij}^t = 0$.
2. For every $(i, j) \in S^*$, by (39) and the element-wise beta-min condition, we derive that

$$|\tilde{\beta}_{ij}^t| \geq |\beta_{ij}^*| - \|\tilde{\beta}^t - \beta^*\|_\infty > 3\mu/2 > 0,$$

which proves that the whole support set S^* is recovered with sign consistency.

Therefore, we prove that with probability greater than $1 - O\left(e^{-\frac{1}{3}[\Delta(1, s_0) + \log(ss_0)]}\right)$, the true support set S^* can be exactly recovered without any falsely discovered entry or group for all $t > C \log n$.

We next show a sharper estimation rate by using Theorem 2.1 in [17], that is, for all $u > 0$,

$$\begin{aligned} & P\left(\left\|(X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top \xi\right\|_2^2 \right. \\ & \quad \geq \text{tr}((X_{S^*}^\top X_{S^*})^{-1}) + 2\sqrt{\text{tr}((X_{S^*}^\top X_{S^*})^{-1}(X_{S^*}^\top X_{S^*})^{-1})} u \\ & \quad \left. + 2\|(X_{S^*}^\top X_{S^*})^{-1}\|_2 u\right) \leq e^{-u}. \end{aligned}$$

Based on DSRIP(s, s_0, δ) condition, we have $\Lambda_i((X_{S^*}^\top X_{S^*})^{-1}) \leq \frac{1}{n(1-\delta)}$, which leads

$$\begin{aligned} \text{tr}((X_{S^*}^\top X_{S^*})^{-1}) & \leq \frac{ss_0}{n(1-\delta)}, \\ \text{tr}((X_{S^*}^\top X_{S^*})^{-1}(X_{S^*}^\top X_{S^*})^{-1}) & \leq \frac{ss_0}{n^2(1-\delta)^2}, \\ \|(X_{S^*}^\top X_{S^*})^{-1}\|_2 & \leq \frac{1}{n(1-\delta)}. \end{aligned}$$

Therefore, for every $\epsilon \in (0, 1)$, by taking $u = \log(1/\epsilon)$, we have

$$P\left(\frac{1}{\sigma^2} \|\tilde{\beta}^* - \beta^*\|_2^2 \leq \frac{2ss_0}{n(1-\delta)} + \frac{3\log(1/\epsilon)}{n(1-\delta)}\right) \geq 1 - \epsilon,$$

combining with (38) we complete the proof of Theorem 2.2.

A.4. Proof of Theorem 2.3

By Cramér–Wold Theorem, it is sufficient to prove the asymptotic normality for $\sqrt{n}g^\top(\tilde{\beta}_{S^*}^t - \beta_{S^*}^*)$, where $g \in \mathbb{R}^{|S^*| \times 1}$ is an arbitrary vector with bounded Euclidean norm. By taking $t > C_\delta \log n$, the output $\tilde{\beta}^t$ of Algorithm 2 leads that

$$\begin{aligned} |\sqrt{n}g^\top(\tilde{\beta}_{S^*}^t - \beta_{S^*}^*) - \sqrt{n}g^\top(\tilde{\beta}_{S^*}^* - \beta_{S^*}^*)| & \leq \sqrt{n}\|g\|_2 \cdot \|\tilde{\beta}_{S^*}^t - \tilde{\beta}_{S^*}^*\|_2 \\ & \leq \|g\|_2 \frac{\sigma}{\sqrt{n}} \rightarrow 0, \end{aligned}$$

as $n, p(=d \times m) \rightarrow \infty$, where the last inequality follows from Step 4 in the proof of Theorem 2.2. Therefore we only need to focus on the term $\sqrt{n}g^\top(\tilde{\beta}_{S^*}^* - \beta_{S^*}^*)$. By definition, we get

$$\sqrt{n}g^\top(\tilde{\beta}_{S^*}^* - \beta_{S^*}^*) = \sum_{k=1}^n \sigma \sqrt{n} \cdot g^\top (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^{(k)} \xi_k := \sum_{k=1}^n \zeta_k, \quad (40)$$

where we denote by $X_{S^*}^{(k)} \in \mathbb{R}^{|S^*| \times 1}$ the k -th observation of the covariates on the support S^* .

Additionally, by the DSRIP condition, we have

$$\sum_{k=1}^n \text{Var}(\zeta_k) = \sigma^2 n c_\xi g^\top (X_{S^*}^\top X_{S^*})^{-1} g \geq \frac{\sigma^2 c_\xi \|g\|_2^2}{1 + \delta}, \quad (41)$$

where recall we define $c_\xi := \text{Var}(\xi_k)$. We also have

$$\begin{aligned} \sum_{k=1}^n \mathbf{E}(|\zeta_k|^3) &= \sigma^3 n^{3/2} \sum_{k=1}^n \left| g^\top (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^{(k)} \right|^3 \mathbf{E}(|\xi_k|^3) \\ &\stackrel{(i)}{\leq} C_1 \sigma^3 n^{3/2} \|g\|_2^3 \|(X_{S^*}^\top X_{S^*})^{-1}\|_2^3 \cdot \sum_{k=1}^n \|X_{S^*}^{(k)}\|_2^3 \\ &\leq \frac{C_1 \sigma^3 \|g\|_2^3 B_{S^*}^3}{n^{1/2}(1 - \delta)^3}, \end{aligned} \quad (42)$$

where inequality (i) follows from the sub-Gaussian property, i.e.,

$$\mathbf{E}|\xi_k|^3 = \int_{t \geq 0} \mathbf{P}(|\xi_k|^3 \geq t) dt \leq \int_{t \geq 0} 6t^2 e^{-t^2/2} dt = 3\sqrt{2\pi} =: C_1,$$

and the last inequality follows from the definition $B_{S^*} = \max_{i \in [n]} \|X_{S^*}^{(i)}\|_2$.

Therefore, we have

$$\begin{aligned} \frac{1}{(\sum_{k=1}^n \text{Var}(\zeta_k))^{3/2}} \sum_{k=1}^n \mathbf{E}(|\zeta_k|^3) &\leq \frac{C_1 \sigma^3 \|g\|_2^3 B_{S^*}^3 (1 + \delta)^{3/2}}{n^{1/2}(1 - \delta)^3 \sigma^3 c_\xi^{3/2} \|g\|_2^3} \\ &= \frac{C_1 (1 + \delta)^{3/2}}{c_\xi^{3/2} (1 - \delta)^3} \cdot \frac{B_{S^*}^3}{\sqrt{n}} \rightarrow 0, \end{aligned} \quad (43)$$

as $B_{S^*}^3 = o(\sqrt{n})$ and $n \rightarrow \infty$. Finally, by Lyapunov's central limit theorem, we conclude that

$$\frac{\sqrt{n}g^\top(\tilde{\beta}_{S^*}^* - \beta_{S^*}^*)}{\sqrt{\sigma^2 c_\xi g^\top \left(\frac{1}{n} X_{S^*}^\top X_{S^*} \right)^{-1} g}} \rightarrow N(0, 1),$$

as $n, d, m \rightarrow \infty$ and $B_{S^*}^3 = o_p(\sqrt{n})$, which completes the proof of Theorem 2.3.

A.5. Proof of Theorem 2.1

The proof of Theorem 2.1 is quite similar to Proposition 2.1, with fewer scale techniques. We first introduce some abbreviations used in this subsection:

$$C_\lambda = \sqrt{40} \cdot \frac{\kappa + (\sqrt{3} - 1)\delta}{\kappa - \delta}, \quad A = \frac{8\delta^2}{(\kappa - \delta)^2}, \quad B := \left(1 - \sqrt{10}/C_\lambda\right)^2 \cdot \frac{8\kappa^2}{(\kappa - \delta)^2}, \quad (44)$$

where recall $\delta \in (0, 1)$ is the parameter in $\text{DSRIP}\left((1 + 2A)s, \frac{1+4A}{1+2A}s_0, \delta\right)$ condition, and $\kappa \in \left(\delta, 1 \wedge (\delta + 2\sqrt{2}\delta)\right)$ is a tuning parameter. We then introduce the event

$$\mathcal{E}_{2.1} := \left\{ \max_{S \in \mathcal{S}(s, As_0) \cup \mathcal{S}(As, s_0)} \sum_{(i,j) \in S} \Xi_{ij}^2 < 10A \cdot \frac{\sigma^2 s s_0 \Delta(s, s_0)}{n} \right\}.$$

Folowing Lemma C.1, we learn that $\mathbf{P}(\mathcal{E}_{2.1}) \geq 1 - 2 \exp\left(-\frac{A}{3} s s_0 \Delta(s, s_0)\right)$.

Under this event, we next prove that

$$\begin{aligned} \|\hat{\beta}^t - \beta^*\|_2 &\leq \sqrt{B s s_0} \lambda_t, \\ \hat{S}_{OG}^t &:= (S_{G^*})^c \cap \hat{S}^t \in \mathcal{S}(As, s_0), \\ \hat{S}_{IG}^t &:= S_{G^*} \cap \hat{S}^t \cap (S^*)^c \in \mathcal{S}(s, As_0), \end{aligned} \quad (45)$$

where $\hat{S}_{OG}^t, \hat{S}_{IG}^t$ represent similar meaning as $\tilde{S}_{OG}^t$ and $\tilde{S}_{IG}^t$ in Table 3 (the superscript $\wedge$ denotes the analysis of the first-stage algorithm, and the superscript $\sim$ denotes the analysis of the second-stage algorithm).

When $t = 0$, we have $\hat{\beta}^0 = \mathbf{0}_p$ and $\hat{S}^0 = \emptyset$, therefore all three conclusions in (45) hold by choosing proper $\lambda_{(0)}$ satisfying $\|\beta^*\|_2 \leq \frac{3(1+\sqrt{2})}{2} \sqrt{s s_0} \lambda_{(0)}$ (which will be discussed in the last of this subsection). Then, by using mathematical induction, we first assume all three results hold in the t -th iteration (for some $t \geq 0$), and aim to prove that all of them still hold in the $(t + 1)$ -th iteration.

Step 1 (control the shape of $\hat{S}_{OG}^{t+1} = (S_{G^*})^c \cap \hat{S}^{t+1}$) By contradiction, we initially assume that more than As groups are falsely discovered in the $(t + 1)$ -th iteration. Then, we can choose arbitrary As false discovered groups and construct an exact (As, s_0) -shape $S'_{OG} \in \mathcal{S}(As, s_0)$ (the process of constructing S'_{OG} is the same as the step 1 in the proof of Proposition 2.1). Based on the element-wise thresholding and the group-wise thresholding operators, for any group G_j selected in S'_{OG} , it holds that $\|\hat{\beta}_{G_j \cap S'_{OG}}^{t+1}\|_2^2 \geq s_0 \lambda_{(t)}^2$, which yields that

$$\begin{aligned} \sqrt{A s s_0} \lambda_{(t+1)} &\leq \sqrt{\sum_{(i,j) \in S'_{OG}} \langle \Phi_{(i,j)}^\top, \beta^* - \hat{\beta}^t \rangle^2} + \sqrt{\sum_{(i,j) \in S'_{OG}} \Xi_{ij}^2} \\ &\stackrel{(i)}{\leq} \delta \|\hat{\beta}^t - \beta^*\|_2 + \sqrt{\frac{10A \sigma^2 s s_0 \Delta(s, s_0)}{n}} \\ &\stackrel{(ii)}{\leq} \delta \sqrt{B s s_0} \lambda_{(t)} + \frac{\sqrt{10A}}{C_\lambda} \sqrt{s s_0} \lambda_{(\infty)} \\ &\stackrel{(iii)}{\leq} \left(\frac{\delta \sqrt{B}}{\kappa} + \frac{\sqrt{10A}}{C_\lambda} \right) \sqrt{s s_0} \lambda_{(t+1)} \\ &= \sqrt{A s s_0} \lambda_{(t+1)}, \end{aligned} \quad (46)$$

where inequality (i) follows from the $\text{DSRIP}\left((1 + 2A)s, \frac{1+4A}{1+2A}s_0, \delta\right)$ condition and event $\mathcal{E}_{2.1}$, inequality (ii) follows from the induction hypothesis (45) in the t -th iteration, and also follows from the relationship (44). Inequality (iii) follows from $\lambda_{(t+1)} = (\kappa \lambda_{(t)}) \vee \lambda_{\infty}$, and the final

equality follows from the relationship (44) again. Thus, we find an absurdity, demonstrating that only fewer than As groups can be falsely discovered in the $(t+1)$ -th iteration.

Again, by contradiction, we can prove that fewer than Ass_0 entries are falsely discovered in the falsely discovered groups. If not so, we can construct a set $S''_{OG} \subseteq \hat{S}_{OG}^{t+1}$ satisfying $|S''_{OG}| = Ass_0$ and $S''_{OG} \in \mathcal{S}(As, s_0)$, which leads an absurdity as the same as (46) again.

Step 2 (control the shape of $\hat{S}_{IG}^{t+1} = S_{G^*} \cap (S^*)^c \cap \hat{S}^{t+1}$) By contradiction, we initially assume that more than Ass_0 entries are falsely discovered in $S_{G^*} \cap (S^*)^c$. Then we can construct a index set S'_{IG} satisfying $S'_{IG} \subseteq \hat{S}_{IG}^{t+1}$, $S'_{IG} \in \mathcal{S}(s, As_0)$ and $|S'_{IG}| = Ass_0$. Therefore, similar to (46), we can find the absurdity again.

Step 3 (ℓ_2 error inequality of $\hat{\beta}^{t+1}$) Now we prove that the ℓ_2 error bound in the $(t+1)$ -th iteration. Note that

$$\hat{\beta}_{ij}^{t+1} - \beta_{ij}^* = -\hat{H}_{ij}^{t+1} \cdot \mathbf{1} \left((i, j) \notin \hat{S}^{t+1} \right) + \langle \Phi_{(i,j)}^\top, \beta^* - \hat{\beta}^t \rangle + \Xi_{i,j},$$

where

$$\hat{H}_{ij}^{t+1} = \beta_{ij}^* + \langle \Phi_{(i,j)}^\top, \beta^* - \hat{\beta}^t \rangle + \Xi_{i,j}.$$

We then have

$$\begin{aligned} & \|\hat{\beta}^{t+1} - \beta^*\|_2 \\ & \leq \left[\sum_{(i,j) \in S^*} \left\{ \langle \Phi_{(i,j)}^\top, \beta^* - \hat{\beta}^t \rangle + \Xi_{ij} - \hat{H}_{ij}^{t+1} \cdot \mathbf{1} \left((i, j) \notin \hat{S}^{t+1} \right) \right\}^2 \right. \\ & \quad \left. + \sum_{(i,j) \in \hat{S}^{t+1} \setminus S^*} \left\{ \langle \Phi_{(i,j)}^\top, \beta^* - \hat{\beta}^t \rangle + \Xi_{ij} \right\}^2 \right]^{1/2} \\ & \leq \sqrt{\sum_{(i,j) \in \hat{S}^{t+1} \cup S^*} \left\{ \langle \Phi_{(i,j)}^\top, \beta^* - \hat{\beta}^t \rangle + \Xi_{ij} \right\}^2} \\ & \quad + \left\{ \sum_{(i,j) \in S^*} \left(\hat{H}_{ij}^{t+1} \right)^2 \mathbf{1} \left(\left| \hat{H}_{ij}^{t+1} \right| < \lambda_{(t+1)} \right) \right. \\ & \quad \left. + \sum_{(i,j) \in S^*} \left(\hat{H}_{ij}^{t+1} \right)^2 \mathbf{1} \left(\left| \hat{H}_{ij}^{t+1} \right| \geq \lambda_{(t+1)} \right) \right. \\ & \quad \left. \times \mathbf{1} \left(\sum_{k:(k,j) \in S^*} \left(\hat{H}_{kj}^{t+1} \right)^2 \mathbf{1} \left(\left| \hat{H}_{kj}^{t+1} \right| \geq \lambda_{(t+1)} \right) < s_0 \lambda_{(t+1)}^2 \right) \right\}^{1/2} \\ & \leq \delta \|\hat{\beta}^t - \beta^*\|_2 + \sqrt{\sum_{(i,j) \in S^*} \Xi_{ij}^2 + \sum_{(i,j) \in \hat{S}_{IG}^{t+1}} \Xi_{ij}^2 + \sum_{(i,j) \in \hat{S}_{OG}^{t+1}} \Xi_{ij}^2 + 2s s_0 \lambda_{(t+1)}}, \end{aligned} \tag{47}$$

where the second inequality follows from the triangle inequality and the definition of the double sparse thresholding operator $\mathcal{T}_{\lambda_{(t+1)}, s_0}$ in the $(t+1)$ -th iteration. The last inequality follows from the DSRIP $\left((1+2A)s, \frac{1+4A}{1+2A}s_0, \delta \right)$ condition, since by the induction hypothesis (45) and

the first two step proofs, we have $S^* \cup \hat{S}^t \cup \hat{S}^{t+1} \in \mathcal{S} \left((1 + 2A)s, \frac{1+4A}{1+2A}s_0 \right)$. Consequently, by the event $\mathcal{E}_{2,1}$ and (45), we conclude that

$$\begin{aligned} \|\hat{\beta}_{S^*}^{t+1} - \beta_{S^*}^*\|_2 &\leq \delta \|\hat{\beta}^t - \beta^*\|_2 + \sqrt{\frac{30A\sigma^2 s s_0 \Delta(s, s_0)}{n}} + \sqrt{2s s_0} \lambda_{(t+1)} \\ &\leq \left(\frac{\delta \sqrt{B}}{\kappa} + \frac{\sqrt{30A}}{C_\lambda} + \sqrt{2} \right) \sqrt{s s_0} \lambda_{(t+1)} \\ &= \sqrt{B s s_0} \lambda_{(t+1)}, \end{aligned} \quad (48)$$

where the last equality follows from the relationship (44).

Therefore, we prove that all three results in (45) still hold in the $(t + 1)$ -th iteration and complete the mathematical induction.

Step 4 (feasible initial thresholding parameter) We finally end the proof of Theorem 2.1 by providing a suitable initial thresholding parameter

$$\lambda_{(0)} := \frac{\|X^\top Y/n\|_\infty + \sqrt{10\sigma^2(\log p)/n}}{\sqrt{2}\kappa}.$$

By the relationship (44), we learn that $C_\lambda > \sqrt{40}$ and $\sqrt{B} \geq \frac{\sqrt{2}\kappa}{\kappa - \delta} \geq \frac{\sqrt{2}\kappa}{1 - \delta}$. Consequently,

$$\begin{aligned} \sqrt{B s s_0} \lambda_{(0)} &\geq \sqrt{s s_0} \cdot \frac{\sqrt{2}\kappa}{1 - \delta} \cdot \left(\frac{\|X^\top Y/n\|_\infty + \sqrt{10\sigma^2(\log p)/n}}{\sqrt{2}\kappa} \right) \\ &\stackrel{(i)}{\geq} \frac{1}{1 - \delta} \cdot \left(\sqrt{s s_0} \left\| \frac{1}{n} X_{S^*}^\top (X_{S^*} \beta_{S^*}^* + \sigma \xi) \right\|_\infty + \|\Xi_{S^*}\|_2 \right) \\ &\geq \frac{1}{1 - \delta} \cdot \left(\left\| \frac{1}{n} X_{S^*}^\top X_{S^*} \beta_{S^*}^* \right\|_2 - \left\| \frac{\sigma}{n} X_{S^*}^\top \xi \right\|_2 + \|\Xi_{S^*}\|_2 \right) \\ &\geq \frac{1}{1 - \delta} \left\| \frac{1}{n} X_{S^*}^\top X_{S^*} \right\|_2 \cdot \|\beta_{S^*}^*\|_2 \\ &\geq \|\beta_{S^*}^*\|_2, \end{aligned}$$

where inequality (i) follows from Lemma C.1, holding with probability greater than $1 - \exp(-s s_0 \Delta(s, s_0)/3)$, and the last inequality follows from DSRIP condition.

Therefore, combining Steps 1-4, we complete the proof of Theorem 2.1.

A.6. Miscellaneous

This subsection provides an in-depth discussion of two assumptions of the design matrix: the DSRIP condition and the rate $B_{S^*} = \max_{i \in [n]} \|X_{i, S^*}\|_2$.

A.6.1. Connection between DSRIP and double sparse Riesz condition

We first introduce the double sparse Riesz condition.

Definition A.1. We say that the design matrix $X \in \mathbb{R}^{n \times p}$ satisfies the Double Sparse Riesz Condition $DSRC(as, bs_0, C_U, C_L)$ if and only if

$$C_L \|u\|_2^2 \leq \frac{1}{n} \|X_S u\|_2^2 \leq C_U \|u\|_2^2, \text{ for every } S \in \mathcal{S}(as, bs_0) \text{ and } u \in \mathbb{R}^{|S|} \setminus \{\mathbf{0}_{|S|}\},$$

where $C_U \geq C_L > 0$ are two arbitrary constants and $a, b > 0$ depend on C_U and C_L .

The sparse Riesz condition is a well-known structural assumption in sparse regression [4, 44], and by Definition A.1, we extend this condition to the double sparse space: Assume X satisfies $DSRC\left((1+2A)s, \frac{1+4A}{1+2A}s_0, C_U, C_L\right)$, and rewrite the gradient descent (26) with a learning rate γ as

$$\begin{aligned} \tilde{H}^{t+1} &= \tilde{\beta}^t + \frac{\gamma}{n} X^\top (Y - X\tilde{\beta}^t) \\ &= \tilde{\beta}^t + \frac{\gamma}{n} X^\top (X\tilde{\beta}^* + \sigma\tilde{\xi} - X\tilde{\beta}^t) = \tilde{\beta}^* + \left(\frac{\gamma}{n} X^\top X - I_p\right) (\tilde{\beta}^* - \tilde{\beta}^t) + \gamma\tilde{\Xi}. \end{aligned} \quad (49)$$

By solving the system of inequalities in γ and δ

$$\begin{cases} |\gamma C_U - 1| \leq \delta, \\ |\gamma C_L - 1| \leq \delta, \end{cases}$$

we obtain the feasible region of learning rate γ is $(0, 2/C_U)$, leading

$$\left\| \frac{\gamma}{n} X_S^\top X_S - I_p \right\|_2 \leq \underbrace{\max\{|\gamma C_U - 1|, |1 - \gamma C_L|\}}_{=: \delta'} < 1$$

for every set $S \in \mathcal{S}\left((1+2A)s, \frac{1+4A}{1+2A}s_0\right)$. By incorporating the revised decomposition (49), we can rescale inequality (27) exactly as under the original DSRIP condition (with parameter δ'), and thus the proof and conclusions of Theorem 2.1 and Proposition 2.1 remain valid, requiring only minor scaling adjustments involving δ' , γ , C_U , and C_L .

The above analysis establishes the equivalence of DSRIP and DSRC while relaxing our original DSRIP assumption. Fundamentally, both conditions bound the deviation of the sample covariance matrix from the identity matrix I_p , thereby enabling a (Gaussian) location model perspective [9, 29, 23] on the double sparse regression.

A.6.2. Realizability of double sparse Riesz condition

We next consider the realizability of the double sparse Riesz condition in the sub-Gaussian setting: For each $i \in [n]$, we assume the i -th observation $X^{(i)} \stackrel{d}{=} \Sigma^{1/2} Z^{(i)}$, where $\Sigma \in \mathbb{R}^{p \times p}$ is the population covariance matrix and $Z^{(1)}, \dots, Z^{(n)} \in \mathbb{R}^p$ are i.i.d. centered 1-sub-Gaussian random vectors such that $\mathbf{E}(Z^{(i)} Z^{(i)\top}) = I_p$. Assume the Σ has bounded eigenvalues as $C'_L \leq \Lambda_j(\Sigma) \leq C'_U$ for every $j \in [p]$, where $C'_U \geq C'_L > 0$ are two absolute constants.

Proposition A.1. *Under the above sub-Gaussian setting, assume*

$$n \geq \left(\frac{C + \sqrt{2/c}}{1 \wedge (C'_L/2)} \right)^2 \times \left\{ (2A+1)s \log(em/s) + (4A+1)ss_0 \log(ed/s_0) \right\}. \quad (50)$$

Then, with a probability greater than $1 - 2e^{-(2A+1)s \log(em/s) - (4A+1)ss_0 \log(ed/s_0)}$, the design matrix $X \in \mathbb{R}^{n \times p}$ satisfies DSRC $\left((1+2A)s, \frac{1+4A}{1+2A}s_0, C_U, C_L\right)$, where

$$A = \frac{8\delta^2}{(\kappa - \delta)^2}, \quad \delta = \frac{C'_U}{C'_U + C'_L}, \quad \kappa \in (\delta, 1), \quad C_U = C'_U + \frac{C'_L}{2}, \quad C_L = \frac{C'_L}{2},$$

and $C, c > 0$ are two absolute constants.

proof of Proposition A.1. For any set S satisfying $S \in \mathcal{S}\left((1+2A)s, \frac{1+4A}{1+2A}s_0\right)$ and $|S| = (1+4A)ss_0$, by Remark 5.40 in [38], we have

$$\mathbf{P}\left(\left\|\frac{1}{n}X_S^\top X_S - \Sigma_{SS}\right\|_2 > \max(\iota, \iota^2)\right) \leq 2e^{-c\iota^2},$$

where $\iota = C\sqrt{\frac{(1+4A)ss_0}{n}} + \frac{t}{\sqrt{n}}$ and C, c are two fixed positive constants.

Then, by taking $t = \sqrt{\frac{2}{c}} \cdot \sqrt{(2A+1)s \log(em/s) + (4A+1)ss_0 \log(ed/s_0)}$ and following the sample size assumption (50), we get $\iota \vee \iota^2 = \iota \leq C'_L/2$, which yields that

$$\begin{aligned} & \mathbf{P}\left(\max_{S \in \mathcal{S}\left((1+2A)s, \frac{1+4A}{1+2A}s_0\right)} \left\|\frac{1}{n}X_S^\top X_S - \Sigma_{SS}\right\|_2 > \frac{C'_L}{2}\right) \\ & \leq \mathbf{P}\left(\bigcup_{\substack{S \in \mathcal{S}\left((1+2A)s, \frac{1+4A}{1+2A}s_0\right), \\ |S| = (1+4A)ss_0}} \left\{\left\|\frac{1}{n}X_S^\top X_S - \Sigma_{SS}\right\|_2 > \iota \vee \iota^2\right\}\right) \\ & \leq 2 \binom{m}{(1+2A)s} \binom{(1+2A)sd}{(1+4A)ss_0} \exp(-c\iota^2) \\ & \leq 2 \exp\left\{-(2A+1)s \log(em/s) - (4A+1)ss_0 \log(ed/s_0)\right\}, \end{aligned} \tag{51}$$

where the last inequality follows from $\binom{y}{x} \leq (ey/x)^x$ for every $0 < x < y$. Therefore, with a probability greater than $1 - 2e^{-(2A+1)s \log(em/s) - (4A+1)ss_0 \log(ed/s_0)}$, for every $S \in \mathcal{S}\left((1+2A)s, \frac{1+4A}{1+2A}s_0\right)$, we have

$$\left\|\frac{1}{n}X_S^\top X_S\right\|_2 \leq \|\Sigma_{SS}\|_2 + C'_L/2 \leq C'_U + C'_L/2 =: C_U,$$

and

$$\left\|\frac{1}{n}X_S^\top X_S\right\|_2 \geq \|\Sigma_{SS}\|_2 - C'_L/2 \geq C'_L/2 =: C_L.$$

Hence, by choosing learning rate $\gamma = \frac{2}{C_U + C_L}$ and $\delta = \frac{C_U - C_L}{C_U + C_L}$, we complete the proof of Proposition A.1. $\square$

Some examples satisfying DSRC Here, we provide some illustrative examples satisfying DSRC. Consider an exponential-decay Toeplitz covariance matrix Σ with entries $\Sigma_{ij} = \rho^{|i-j|}$ for a constant $\rho \in [0, 1)$. This covariance structure is ubiquitous in high-dimensional studies [32, 15, 47]. It can be proved that

$$\Lambda_k(\Sigma) = \frac{1 - \rho^2}{1 - 2\rho \cos(\frac{\pi k}{p+1}) + \rho^2}, \text{ for every } k \in [p],$$

leading

$$C'_L = \frac{1 - \rho}{1 + \rho} \leq \Lambda_k(\Sigma) \leq \frac{1 + \rho}{1 - \rho} = C'_U.$$

Therefore, this Σ satisfies our DSRC assumption.

Additionally, consider the special case of an i.i.d. Gaussian design $X_{ij} \sim N(0, 1)$ (so $\rho = 0$ and $C'_U = C'_L = 1$), which is standard in compressed sensing [31] and high-dimensional sparse regression [33]. Then the design matrix X even satisfies the DSRIP condition with high probability as soon as $n \gtrsim s \log(em/s) + ss_0 \log(ed/s_0)$.

A.6.3. A high probability bound of B_{S^*}

Here we provide the rate of B_{S^*} (appears in Theorem 2.3) in the context of sub-Gaussian design. For each $i \in [n]$, we consider the vector $X^{(i)} \stackrel{d}{=} \Sigma^{1/2} Z^{(i)} \in \mathbb{R}^p$ as in Appendix A.6.2, and assume the submatrix $\Sigma_{S^*, S^*} \in \mathbb{R}^{|S^*| \times |S^*|}$ has bounded spectral norm as $\Lambda_{\max}(\Sigma_{S^*, S^*}) \leq C_{S^*}$, where C_{S^*} is an absolute constant. Then, by the Hanson-Wright inequality [34], for a fixed $i \in [n]$, we have

$$\mathbf{P} \left\{ \left\| X_{S^*}^{(i)} \right\|_2^2 \geq \text{tr}(\Sigma_{S^*, S^*}) + K^2 \left(\sqrt{\frac{t}{c}} \|\Sigma_{S^*, S^*}\|_F + \frac{t}{c} \|\Sigma_{S^*, S^*}\|_2 \right) \right\} \leq e^{-t}$$

with two constants $K > 0, c \in (0, 1)$. We take $t = 2 \log n$, by the inequalities

$$\begin{aligned} \text{tr}(\Sigma_{S^*, S^*}) + K^2 \left(\sqrt{\frac{t}{c}} \|\Sigma_{S^*, S^*}\|_F + \frac{t}{c} \|\Sigma_{S^*, S^*}\|_2 \right) &\leq C_{S^*} s s_0 + \frac{C_{S^*} K^2}{c} (\sqrt{s s_0 t} + t) \\ &\leq \frac{3}{2} C_{S^*} \left(1 + \frac{K^2}{c} \right) (s s_0 + t), \end{aligned}$$

we have

$$\begin{aligned} &\mathbf{P} \left[\bigcup_{i \in [n]} \left\{ \left\| X_{S^*}^{(i)} \right\|_2^2 \geq \frac{3}{2} C_{S^*} \left(1 + \frac{K^2}{c} \right) (s s_0 + 2 \log n) \right\} \right] \\ &\leq \sum_{i \in [n]} \mathbf{P} \left\{ \left\| X_{S^*}^{(i)} \right\|_2^2 \geq \frac{3}{2} C_{S^*} \left(1 + \frac{K^2}{c} \right) (s s_0 + 2 \log n) \right\} \\ &\leq n e^{-2 \log n} = n^{-1}. \end{aligned} \tag{52}$$

With a probability greater than $1 - 1/n$, (52) leads $B_{S^*} \lesssim \sqrt{s s_0 + \log n}$. Since $\log n \prec n^{1/3}$ as $n \rightarrow \infty$, it follows that

$$\{B_{S^*}^3 = o_p(\sqrt{n})\} \Leftrightarrow \{(s s_0)^3 + \log^3 n = o_p(n)\} \Leftrightarrow \{s s_0 = o_p(n^{1/3})\},$$

which provides a clear understanding of the technical assumption in Theorem 2.3.

Appendix B: Proof of the minimax lower bounds

First, we consider the minimax lower bound for signal estimation and transform the minimax risk into Bayesian risk for in-depth analysis. This analytical framework is inspired by [29].

Lemma B.1. *For any $1 \leq s < m$, $1 \leq s_0 < d$, any subset $\Theta \subseteq \mathbb{R}^P$ and for any prior probability distribution π on $\mathbb{R}^P$, as we assume the design matrix $X \in \mathbb{R}^{n \times P}$ is fixed, we have*

$$\inf_{\hat{\beta}} \sup_{\beta^* \in \Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\beta} - \beta^*\|_2^2 \geq \inf_{\hat{T}} \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}(Y, X) - \beta^*\|_2^2 - 2 \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \left(\mathbf{E}_{\beta^\Theta | Y} \left(\|\beta^\Theta\|_2^2 | Y \right) + \|\beta^*\|_2^2 \right) \mathbf{1}(\beta^* \notin \Theta) \right\}, \quad (53)$$

where $\inf_{\hat{\beta}}, \inf_{\hat{T}}$ are taken over all estimator of β^* , and $\beta^\Theta := \beta \mathbf{1}(\beta \in \Theta)$. We denote by $\mathbf{E}_{\beta^\Theta | Y}(\cdot | Y)$ an expectation based on the conditional distribution $\frac{P(Y|\beta)\pi_\Theta(\beta)}{\int P(Y|\beta)\pi_\Theta(\beta)d\beta}$, where π_Θ denotes the probability measure π conditioned by the event $\{\beta \in \Theta\}$.

B.1. Proof of Theorem 3.1

Recall that we mainly focus on two subsets of $\Theta_e(s, s_0, a)$, which can be described as

$$\begin{aligned} \Theta_{e,1} &:= \left\{ \beta \in \Theta_e(s, s_0, a) \left| \begin{array}{l} G^*(\beta) = [s], \\ \beta_{ij} = a \text{ for every } (i, j) \in \text{supp}(\beta) \end{array} \right. \right\}, \\ \Theta_{e,2} &:= \left\{ \beta \in \Theta_e(s, s_0, a) \left| \beta_{ij} = \begin{cases} a, & \text{if } i \in [s_0] \text{ and } j \in G^*(\beta) \\ 0, & \text{otherwise} \end{cases} \right. \right\}. \end{aligned}$$

In $\Theta_{e,1}$, information is limited to the location of support groups, omitting details on their support entries. The subspace $\Theta_{e,2}$ provides insight into the support entries within each support group, but it lacks information on the location of support groups. Given the settings above, we frame the signal estimation as a support identification problem, i.e., to estimate the decoder $\eta^* = \{\mathbf{1}(\beta_{ij}^* \neq 0)\}_{ij} \in \{0, 1\}^P$. Plus, we define two functions:

$$\begin{aligned} \psi(d, s_0, a, \sigma) &:= (d - s_0)\Phi\left(-\frac{t(a, d, s_0, \sigma)}{\sigma}\right) + s_0\Phi\left(-\frac{a\sqrt{n} - t(a, d, s_0, \sigma)}{\sigma}\right), \\ t(a, d, s_0, \sigma) &:= \frac{a\sqrt{n}}{2} + \frac{\sigma^2}{a\sqrt{n}} \log \frac{d - s_0}{s_0}. \end{aligned} \quad (54)$$

Now, we define two prior distributions π_1 and π_2 corresponding to parameter subsets $\Theta_{e,1}$ and $\Theta_{e,2}$ respectively. For π_1 , suppose that for each $(i, j) \in [d] \times [s]$ (by the definition of $\Theta_{e,1}$, the support index must be in $[d] \times [s]$), the number of support entries is from a binomial distribution, i.e., $\sum_{(i,j) \in [d] \times [s]} \eta_{ij}^* \sim \text{Bin}(ds, s'_0/d)$, where s'_0 (satisfies $1 \leq s'_0 < s_0$) is an integer that will be determined later. Assume $\eta_{G_j}^* \equiv \mathbf{0}_d$ for all $j > s$.

For π_2 , suppose that the number of support groups is from a binomial distribution $\text{Bin}(m, s'/m)$, where $s' < s$ is also an integer that will be determined later. In each support group, only the first s_0 entries are support entries. Therefore $\eta_{ij}^* \equiv 0$ for all $i > s_0, j \in [m]$.

Note that π_1 and π_2 are two prior distributions of η^* , while $\beta^* = a \cdot \eta^*$ in $\Theta_{e,1}$ and $\Theta_{e,2}$. Therefore, the term $a\hat{\eta}$ can also be an estimator of β^* , and we replace β^* and $\hat{\beta}$ (in Lemma B.1) by η^* and $\hat{\eta}$ with prior π_k (for $k = 1, 2$). Then we have

$$\begin{aligned}
& \inf_{\hat{\eta} \in \{0,1\}^p} \sup_{\beta^* \in \Theta_{e,k}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta} - \eta^*\|_2^2 \\
& \geq \inf_{\hat{\eta} \in [0,1]^p} \sup_{\beta^* \in \Theta_{e,k}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta} - \eta^*\|_2^2 \\
& \geq \inf_{\hat{T}} \mathbf{E}_{\eta^* \sim \pi_k} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}(Y, X) - \eta^*\|_2^2 \\
& \quad - 2 \mathbf{E}_{\eta^* \sim \pi_k} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \left(\mathbf{E}_{\eta^{\Theta_{e,k}} | Y} (\|\eta^{\Theta_{e,k}}\|_2^2 | Y) + \|\eta^*\|_2^2 \right) \mathbf{1}(a\eta^* \notin \Theta_{e,k}) \right\} \\
& \geq \inf_{\hat{T}} \mathbf{E}_{\eta^* \sim \pi_k} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}(Y, X) - \eta^*\|_2^2 \\
& \quad - 2ss_0 \mathbf{P}_{\eta^* \sim \pi_k} (a\eta^* \notin \Theta_{e,k}) - 2 \mathbf{E}_{\eta^* \sim \pi_k} (\|\eta^*\|_2^2 \mathbf{1}(a\eta^* \notin \Theta_{e,k})),
\end{aligned} \tag{55}$$

where $\inf_{\hat{T}}$ denotes the infimum over all estimators $\hat{T} \in [0, 1]^p$, and the last inequality follows from $\|\eta^{\Theta_{e,k}}\|_2^2 = \|\eta\|_2^2 \cdot \mathbf{1}(a\eta \in \Theta_{e,k}) \leq ss_0$.

Proof of (15). Recall that by $\beta^* \in \Theta_{e,1}$, only the first s groups are support groups. And based on the prior π_1 , the total number of support entries $v := \sum_{j \in [m]} v_j$ is from a binomial distribution $\text{Bin}(ds, s'_0/d)$, whence we have

$$\begin{aligned}
& 2ss_0 \mathbf{P}_{\eta^* \sim \pi_1} (a\eta^* \notin \Theta_{e,1}) + 2\mathbf{E}_{\eta^* \sim \pi_1} (\|\eta^*\|_2^2 \mathbf{1}(a\eta^* \notin \Theta_{e,1})) \\
& = 2ss_0 \mathbf{P}(v > ss_0) + 2\mathbf{E}\{v \cdot \mathbf{1}(v > ss_0)\} \\
& \leq 2ss_0 \exp\left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)}\right) + 2ss'_0 \exp\left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)}\right),
\end{aligned} \tag{56}$$

where the first equality follows from the definition of Θ_1 , and the last inequality follows from Lemma D.1 and the bound

$$\begin{aligned}
\mathbf{E}\{v \cdot \mathbf{1}(v > ss_0)\} &= \sum_{i=ss_0+1}^{ds} i \binom{ds}{i} \left(\frac{s'_0}{d}\right)^i \left(1 - \frac{s'_0}{d}\right)^{ds-i} \\
&= ss'_0 \sum_{i-1=ss_0}^{ds-1} \frac{(ds-1)!}{(i-1)! [(ds-1)-(i-1)]!} \left(\frac{s'_0}{d}\right)^{i-1} \left(1 - \frac{s'_0}{d}\right)^{(ds-1)-(i-1)} \\
&= ss'_0 P\left(\text{Bin}(ds-1, \frac{s'_0}{d}) \geq ss_0\right) \\
&\leq ss'_0 P\left(\text{Bin}(ds, \frac{s'_0}{d}) \geq ss_0\right) \leq ss'_0 \exp\left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)}\right).
\end{aligned}$$

For the first term in (55), define

$$\tilde{Y}_{ij} := Y - \sum_{(k,\ell) \neq (i,j)} X_{(k\ell)} \beta_{k\ell}^* = X_{(ij)} \beta_{ij}^* + \sigma \xi.$$

Then we have

$$\begin{aligned}
\inf_{\hat{T}} \mathbf{E}_{\eta^* \sim \pi_1 Y \sim P_{\beta^*}} \mathbf{E} \|\hat{T}(Y, X) - \eta^*\|_2^2 &\stackrel{(i)}{\geq} \sum_{j=1}^s \sum_{i=1}^d \mathbf{E}_{\eta_{\setminus(ij)}^*} \left\{ \inf_{\hat{T}_{ij}} \mathbf{E}_{\eta_{ij}^* Y} \left((\hat{T}_{ij}(Y, X) - \eta_{ij}^*)^2 \middle| \eta_{\setminus(ij)}^* \right) \right\} \\
&\stackrel{(ii)}{\geq} \sum_{j=1}^s \sum_{i=1}^d \mathbf{E}_{\eta_{\setminus(ij)}^*} \left\{ \inf_{\hat{T}_{ij}} \mathbf{E}_{\eta_{ij}^* Y} \left((\hat{T}_{ij}(\tilde{Y}_{ij}, X) - \eta_{ij}^*)^2 \middle| \eta_{\setminus(ij)}^* \right) \right\} \\
&= \sum_{j=1}^s \sum_{i=1}^d \inf_{\hat{T}_{ij}} \left\{ \left(1 - \frac{s'_0}{d} \right) \mathbf{E}_{\tilde{Y}_{ij} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)} (\hat{T}_{ij}(\tilde{Y}_{ij}, X))^2 \right. \\
&\quad \left. + \frac{s'_0}{d} \mathbf{E}_{\tilde{Y}_{ij} \sim N(aX_{ij}, \sigma^2 \mathbf{I}_n)} (1 - \hat{T}_{ij}(\tilde{Y}_{ij}, X))^2 \right\},
\end{aligned}$$

where inequality (i) only involves the first s groups since we can always set $\hat{T}_{ij} = 0$ for all $j > s, i \in [d]$ by the construction of π_1 . Inequality (ii) is based on the fact that, under independent prior distributions of the entries of η^* , the oracle selector of a given component η_{ij}^* does not depend on the rest of the components. And the infimum in the last equality can be achieved by the selector

$$\hat{T}_{ij}^*(\tilde{Y}_{ij}, X) := \frac{1}{1 + \frac{(d-s'_0)\varphi_{\sigma}(\tilde{Y}_{ij})}{s'_0\varphi_{\sigma}(\tilde{Y}_{ij}-aX_{ij})}} = \frac{1}{1 + \frac{d-s'_0}{s'_0} \frac{\varphi_{0,\sigma}(\tilde{Y}_{ij})}{\varphi_{a,\sigma}(\tilde{Y}_{ij})}},$$

where we abbreviate $\varphi_{0,\sigma}(\tilde{Y}_{ij}) := \varphi_{\sigma}(\tilde{Y}_{ij})$, $\varphi_{a,\sigma}(\tilde{Y}_{ij}) := \varphi_{\sigma}(\tilde{Y}_{ij} - aX_{ij})$, where $\varphi_{\sigma}(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2}{2\sigma^2}\right)$. Then, define $\mathcal{A} := \left\{ \mathbf{y} \in \mathbb{R}^n : \frac{d-s'_0}{s'_0} \frac{\varphi_{0,\sigma}(\mathbf{y})}{\varphi_{a,\sigma}(\mathbf{y})} > 1 \right\}$, after some simple calculation we have

$$\begin{aligned}
&\inf_{\hat{T} \in [0,1]^p} \mathbf{E}_{\eta^* \sim \pi_1 Y \sim P_{\beta^*}} \mathbf{E} \|\hat{T}(Y, X) - \eta^*\|_2^2 \\
&\geq sd \int_{\mathbf{y} \in \mathcal{A}} \frac{s'_0}{d} \frac{\frac{d-s'_0}{s'_0} \frac{\varphi_{0,\sigma}(\mathbf{y})}{\varphi_{a,\sigma}(\mathbf{y})}}{1 + \frac{d-s'_0}{s'_0} \frac{\varphi_{0,\sigma}(\mathbf{y})}{\varphi_{a,\sigma}(\mathbf{y})}} \varphi_{a,\sigma}(\mathbf{y}) d\mathbf{y} + sd \int_{\mathbf{y} \in \mathcal{A}^c} \frac{s'_0}{d} \frac{\frac{d-s'_0}{s'_0} \frac{\varphi_{0,\sigma}(\mathbf{y})}{\varphi_{a,\sigma}(\mathbf{y})}}{1 + \frac{d-s'_0}{s'_0} \frac{\varphi_{0,\sigma}(\mathbf{y})}{\varphi_{a,\sigma}(\mathbf{y})}} \varphi_{a,\sigma}(\mathbf{y}) d\mathbf{y} \\
&\geq sd \int_{\mathbf{y} \in \mathcal{A}} \frac{s'_0}{d} \cdot \frac{1}{2} \varphi_{a,\sigma}(\mathbf{y}) d\mathbf{y} + sd \int_{\mathbf{y} \in \mathcal{A}^c} \frac{d-s'_0}{d} \cdot \frac{1}{2} \varphi_{0,\sigma}(\mathbf{y}) d\mathbf{y} \\
&= \frac{sd}{2} \left\{ \frac{s'_0}{d} \Phi\left(-\frac{a\sqrt{n}}{2\sigma} + \frac{\sigma}{a\sqrt{n}} \log \frac{d-s'_0}{s'_0}\right) + \frac{d-s'_0}{d} \Phi\left(-\frac{a\sqrt{n}}{2\sigma} - \frac{\sigma}{a\sqrt{n}} \log \frac{d-s'_0}{s'_0}\right) \right\} \\
&= \frac{s}{2} \psi(d, s'_0, a, \sigma),
\end{aligned} \tag{57}$$

where the function ψ follows from (54).

Combining (55), (56) and (57) together, we have

$$\begin{aligned}
&\inf_{\hat{\eta} \in \{0,1\}^p} \sup_{\beta^* \in \Theta_{e,1}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta}(Y, X) - \eta\|_2^2 \\
&\geq \frac{s}{2} \psi(d, s'_0, a, \sigma) - 2s(s_0 + s'_0) \exp\left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)}\right) \\
&\geq \frac{ss'_0}{2s_0} \psi(d, s_0, a, \sigma) - 2s(s_0 + s'_0) \exp\left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)}\right),
\end{aligned} \tag{58}$$

where the last inequality follows from Lemma D.2. $\square$

Proof of (16). Similar to (55), by using the Bayesian risk we have

$$\begin{aligned}
& \inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_{e,2}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta}_G - \eta_G^*\|_2^2 \\
& \geq \inf_{\hat{T}_G \in [0,1]^m} \mathbf{E}_{\eta^* \sim \pi_2} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}_G(Y, X) - \eta_G^*\|_2^2 \\
& \quad - 2 \mathbf{E}_{\eta^* \sim \pi_2} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \left(\mathbf{E}_{\eta^* \in \Theta_{e,2} | Y} \left(\|\eta_G^{\Theta_{e,2}}\|_2^2 | Y \right) + \|\eta_G^*\|_2^2 \right) \mathbf{1}(a\eta^* \notin \Theta_{e,2}) \right\} \\
& \geq \inf_{\hat{T}_G \in [0,1]^m} \mathbf{E}_{\eta^* \sim \pi_2} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}_G(Y, X) - \eta_G^*\|_2^2 \\
& \quad - 2s \mathbf{P}_{\eta^* \sim \pi_2} (a\eta^* \notin \Theta_{e,2}) - 2 \mathbf{E}_{\eta^* \sim \pi_2} (\|\eta_G^*\|_2^2 \mathbf{1}(a\eta^* \notin \Theta_{e,2})),
\end{aligned} \tag{59}$$

where for a $\eta^* \sim \pi_2$, we have $\|\eta_G^*\|_2^2 = \sum_{j \in [m]} (\eta_G^*)_j \sim \text{Bin}(m, s'/m)$. By the construction of π_2 and $\Theta_{e,2}$, similar to (56), we have

$$2s \mathbf{P}_{\eta^* \sim \pi_2} (a\eta^* \notin \Theta_{e,2}) + 2 \mathbf{E}_{\eta^* \sim \pi_2} (\|\eta_G^*\|_2^2 \mathbf{1}(a\eta^* \notin \Theta_{e,2})) \leq 2(s + s') \exp\left(-\frac{3(s - s')^2}{2(s + 2s')}\right).$$

For the first term of (59), we have

$$\begin{aligned}
& \inf_{\hat{T}_G \in [0,1]^m} \mathbf{E}_{\eta^* \sim \pi_2} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}_G(Y, X) - \eta_G^*\|_2^2 \\
& \geq \sum_{j=1}^m \mathbf{E}_{(\eta_G^*)_{\setminus j}} \left\{ \inf_{\hat{T}_{G_j}} \mathbf{E}_{(\eta_G^*)_j | Y} \mathbf{E}_Y \left(|\hat{T}_{G_j}(Y, X) - (\eta_G^*)_j|^2 \middle| (\eta_G^*)_{\setminus j} \right) \right\} \\
& \geq \sum_{j=1}^m \mathbf{E}_{(\eta_G^*)_{\setminus j}} \left\{ \inf_{\hat{T}_{G_j}} \mathbf{E}_{(\eta_G^*)_j | Y} \mathbf{E}_Y \left(|\hat{T}_{G_j}(\tilde{Y}_{G_j}, X) - (\eta_G^*)_j|^2 \middle| (\eta_G^*)_{\setminus j} \right) \right\} \\
& = \sum_{j=1}^m \mathbf{E}_{(\eta_G^*)_{\setminus j}} \left\{ \inf_{\hat{T}_{G_j}} \left[\frac{m - s'}{m} \mathbf{E}_{\tilde{Y}_{G_j} | (\eta_G^*)_j=0} \left(\hat{T}_{G_j}^2 \middle| (\eta_G^*)_{\setminus j} \right) + \frac{s'}{m} \mathbf{E}_{\tilde{Y}_{G_j} | (\eta_G^*)_j=1} \left((1 - \hat{T}_{G_j})^2 \middle| (\eta_G^*)_{\setminus j} \right) \right] \right\} \\
& = \sum_{j=1}^m \mathbf{E}_{(\eta_G^*)_{\setminus j}} \left\{ \frac{m - s'}{m} \mathbf{E}_{\tilde{Y}_{G_j} | (\eta_G^*)_j=0} \left((\hat{T}_{G_j}^*)^2 \middle| (\eta_G^*)_{\setminus j} \right) + \frac{s'}{m} \mathbf{E}_{\tilde{Y}_{G_j} | (\eta_G^*)_j=1} \left((1 - \hat{T}_{G_j}^*)^2 \middle| (\eta_G^*)_{\setminus j} \right) \right\},
\end{aligned}$$

where in the second inequality, we use

$$\tilde{Y}_{G_j} := Y - \sum_{k \neq j} \sum_{i \in [s_0]} \beta_{ik}^* X_{(ik)} = \sum_{i \in [s_0]} a\eta_{ij}^* X_{(ij)} + \sigma \xi \in \mathbb{R}^n$$

to represent the marginal observation under the information of the other groups. And in the last equality, we achieve the infimum by using the selector

$$\hat{T}_{G_j}^*(\tilde{Y}_{G_j}, X) := \frac{1}{1 + \frac{m-s'}{s'} \frac{\varphi_{\sigma}(\tilde{Y}_{G_j})}{\varphi_{\sigma}(\tilde{Y}_{G_j} - \sum_{i \in [s_0]} aX_{(ij)})}}.$$

Therefore, leveraging a technique similar to (57), we obtain

$$\begin{aligned}
& \inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_{e,2}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta}_G(Y, X) - \eta_G^*\|_2^2 \\
& \geq \frac{s'}{2} \left\{ \frac{m-s'}{s'} \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} - \frac{\sigma \log(m/s' - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right. \\
& \quad \left. + \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} + \frac{\sigma \log(m/s' - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right\} \\
& \geq \frac{s'}{2} \left\{ \frac{m-s}{s} \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} - \frac{\sigma \log(m/s - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right. \\
& \quad \left. + \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} + \frac{\sigma \log(m/s - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right\}, \tag{60}
\end{aligned}$$

where the last inequality follows Lemma D.2.

Combining (59) and (60) together, we have

$$\begin{aligned}
& \inf_{\hat{\eta}} \sup_{\beta \in \Theta_{e,2}} \mathbf{E}_{Y \sim P_{\beta}} \|\hat{\eta}(Y, X) - \eta\|_2^2 \\
& \geq \frac{s'}{2s} \left\{ (m-s) \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} - \frac{\sigma \log(m/s - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right. \\
& \quad \left. + s \Phi \left(-\frac{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2}{2\sigma} + \frac{\sigma \log(m/s - 1)}{a \|\sum_{i \in [s_0]} X_{(ij)}\|_2} \right) \right\} - 2(s+s') \exp \left(-\frac{3(s-s')^2}{2(s+2s')} \right). \tag{61}
\end{aligned}$$

Therefore, we complete the proof of Theorem 3.1. $\square$

B.2. Proof of Theorem 3.2

We next prove (18) and (19), respectively.

Proof of (18) Assume $ss_0 \geq 54$ and $a^2 \leq \frac{\sigma^2}{5n} \log(ds - ss_0)$. We stress that the constants appearing in this proof may not be optimal. Regardless, the existence of these lower bounds is assured. Define $B := \frac{\log(ss_0)}{10} + \log(d/s_0 - 1)$, by $a \leq \frac{\sqrt{2}\sigma}{\sqrt{n}} \sqrt{B + \sqrt{B^2 - \log^2(d/s_0 - 1)}}$, we can calculate that

$$\begin{aligned}
& -\frac{a\sqrt{n}}{2\sigma} + \frac{\sigma \log(d/s_0 - 1)}{a\sqrt{n}} \\
& \geq \frac{\sqrt{B - \sqrt{B^2 - \log^2(d/s_0 - 1)}} - \sqrt{B + \sqrt{B^2 - \log^2(d/s_0 - 1)}}}{\sqrt{2}} \\
& = -\sqrt{B - \log(d/s_0 - 1)} \\
& = -\sqrt{\frac{\log(ss_0)}{10}}, \tag{62}
\end{aligned}$$

where the first inequality holds because the function $f(x) = -\frac{x}{2\sigma} + \frac{\sigma \log(d/s_0 - 1)}{x}$ is monotonically decreasing for $x > 0$.

Take $s'_0 = s_0/2$. Then, by combining (62) with (15), we have

$$\begin{aligned}
& \inf_{\hat{\eta} \in \{0,1\}^P} \sup_{\beta^* \in \Theta_{e,1}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta}(Y, X) - \eta^*\|_2^2 \\
& \geq \frac{ss_0}{4} \Phi \left(-\frac{a\sqrt{n}}{2\sigma} + \frac{\sigma \log(d/s_0 - 1)}{a\sqrt{n}} \right) - 3ss_0 \exp \left(-\frac{3ss_0}{16} \right) \\
& \stackrel{(i)}{\geq} \frac{(ss_0)^{19/20}}{8} \frac{\sqrt{2/\pi}}{1 + \sqrt{(\log(ss_0))/10}} - 3(ss_0)^{4/5} \cdot (ss_0)^{1/5} \exp \left(-\frac{3ss_0}{16} \right) \\
& \stackrel{(ii)}{\geq} \frac{1}{9} (ss_0)^{\frac{4}{5}} - \frac{1}{90} (ss_0)^{\frac{4}{5}} \\
& = \frac{1}{10} (ss_0)^{\frac{4}{5}},
\end{aligned} \tag{63}$$

where inequality (i) follows from (62) and $\Phi(-y) \geq \sqrt{\frac{2}{\pi}} \frac{\exp(-y^2/2)}{2(y+1)}$ for every $y > 0$, and inequality (ii) follows from the assumption $ss_0 \geq 54$.

Proof of (19) Using a technique similar to (62), we can also demonstrate that, if $a^2 s_0 \leq \frac{\sigma^2 \log(m-s)}{5(1+\delta)n}$, then

$$\begin{aligned}
& \inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_{e,2}} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{j \in [m]} |(\hat{\eta}_G)_j(Y, X) - (\eta_G^*)_j| \right\} \\
& \geq \frac{s}{4} \Phi \left(-\sqrt{\frac{\log s}{10}} \right) - 3s \exp \left(-\frac{3s}{16} \right) \\
& \geq \frac{1}{8} s^{\frac{7}{10}} - s^{\frac{7}{10}} \times 3s^{\frac{3}{10}} e^{-3s/16} \\
& \geq \frac{1}{20} s^{\frac{7}{10}},
\end{aligned} \tag{64}$$

where the last two inequalities follow from $s \geq 25$.

Combining the results derived from $\Theta_{e,1}$ and $\Theta_{e,2}$, if

$$a^2 \leq \frac{\sigma^2}{10n} \left(\frac{\log(m-s)}{s_0(1+\delta)} + \log(sd - ss_0) \right) \leq \max \left(\frac{\sigma^2 \log(m-s)}{5s_0n(1+\delta)}, \frac{\sigma^2 \log(sd - ss_0)}{5n} \right),$$

then it is impossible for both element-wise and group-wise selection to be consistent—at least one level of selection is unattainable. Therefore, we conclude the proof of Theorem 3.2.

B.3. Proof of Theorem 3.3

Preliminary The group-wise signal analysis is quite similar to the proof of Theorem 3.1 and 3.2. We construct two subspaces of $\Theta_g(s, s_0, b)$ as

$$\begin{aligned}
\Theta_{g,1} &:= \left\{ \beta \in \Theta(s, s_0) \left| \begin{array}{l} G^*(\beta) = [s], \\ \|\beta_{G_j}\|_0 \geq s_0/10 \text{ for every } j \in [s], \\ \beta_{ij} = a \text{ for every } (i, j) \in \text{supp}(\beta) \end{array} \right. \right\}, \\
\Theta_{g,2} &:= \left\{ \beta \in \Theta(s, s_0) \left| \beta_{ij} = \begin{cases} b/\sqrt{s_0}, & \text{if } i \in [s_0] \text{ and } j \in G^*(\beta) \\ 0, & \text{otherwise} \end{cases} \right. \right\}.
\end{aligned}$$

where in $\Theta_{g,1}$, the parameter a satisfies that $\|\beta_{G_j}\|_2 = \sqrt{a^2 \|\beta_{G_j}\|_0} \geq b$ for every $j \in [s]$. The proof of Theorem 3.3 uses Lemma B.1 with the priors π_1 and π_2 defined in Appendix B.1.

Proof of (23) We first concentrate on the lower bound for group-wise support recovery over $\Theta_{g,2}$. Since $\Theta_{g,2}$ and $\Theta_{e,2}$ share the same structural properties, the arguments in (59), (60), (61), and (64) carry over directly. In particular, taking $s' = s/2 \geq 12.5$ and $b^2 \leq \frac{\sigma^2 \log(m-s)}{5(1+\delta)n}$, we obtain

$$\inf_{\hat{\eta}_G \in \{0,1\}^m} \sup_{\beta^* \in \Theta_{g,2}} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{j \in [m]} |(\hat{\eta}_G)_j(Y, X) - (\eta_G^*)_j| \right\} \geq \frac{1}{20} s^{\frac{7}{10}}.$$

Proof of (22) We then focus on the element-wise selection error in $\Theta_{g,1}$. With the prior π_1 and $1 \leq s'_0 \leq s_0$, we have

$$\begin{aligned} & \inf_{\hat{\eta} \in \{0,1\}^{d \times m}} \sup_{\beta^* \in \Theta_{g,1}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta} - \eta^*\|_2^2 \\ & \geq \inf_{\hat{T}} \mathbf{E}_{\eta^* \sim \pi_1} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}(Y, X) - \eta^*\|_2^2 - 2ss_0 \mathbf{P}_{\eta^* \sim \pi_1} (a\eta^* \notin \Theta_{g,1}) - 2 \mathbf{E}_{\eta^* \sim \pi_1} (\|\eta^*\|_2^2 \mathbf{1}(a\eta^* \notin \Theta_{g,1})) \\ & \geq \frac{ss'_0}{2s_0} \left\{ (d - s_0) \Phi \left(-\frac{a\sqrt{n}}{2\sigma} - \frac{\sigma \log(d/s_0 - 1)}{a\sqrt{n}} \right) + s_0 \Phi \left(-\frac{a\sqrt{n}}{2\sigma} + \frac{\sigma \log(d/s_0 - 1)}{a\sqrt{n}} \right) \right\} \\ & \quad - 2s(s_0 + s'_0) \exp \left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)} \right) - 4s^2 s_0 \exp \left(-\frac{15(s'_0 - s_0/10)^2}{40s'_0 - s_0} \right), \end{aligned} \quad (65)$$

where the first inequality follows from (55), and in the last inequality, the first term (Bayesian risk) is derived from (57) and Lemma D.2, and the last two terms are from the following:

$$\begin{aligned} & 2ss_0 \mathbf{P}_{\eta^* \sim \pi_1} (a\eta^* \notin \Theta_{g,1}) + 2 \mathbf{E}_{\eta^* \sim \pi_1} (\|\eta^*\|_2^2 \mathbf{1}(a\eta^* \notin \Theta_{g,1})) \\ & = 2ss_0 \left\{ \mathbf{P}(v > ss_0) + \mathbf{P}(v \leq ss_0, \text{ and } v_j < s_0/10 \text{ for some } j \in [s]) \right\} \\ & \quad + 2 \mathbf{E} \left\{ v \left[\mathbf{1}(v > ss_0) + \mathbf{1}(v \leq ss_0, \text{ and } v_j < s_0/10 \text{ for some } j \in [s]) \right] \right\} \\ & \leq 2ss_0 \mathbf{P}(v > ss_0) + 2 \mathbf{E} \{v \mathbf{1}(v > ss_0)\} + 4ss_0 \sum_{j \in [s]} \mathbf{P}(v_j < s_0/10) \\ & \leq 2s(s_0 + s'_0) \exp \left(-\frac{3s(s_0 - s'_0)^2}{2(s_0 + 2s'_0)} \right) + 4s^2 s_0 \exp \left(-\frac{15(s'_0 - s_0/10)^2}{40s'_0 - s_0} \right). \end{aligned}$$

We now take $s'_0 = 2s_0/3$. Following the similar proof technique used in the first part of Appendix B.2, as $a^2 = \frac{\sigma^2 \log(ds - ss_0)}{10n}$ (thus leading $b^2 \leq \frac{\sigma^2 s_0 \log(ds - ss_0)}{100n}$), $ss_0 \geq 87$, and $s^{6/5} \leq \frac{7}{200} s_0^{-1/5} \exp(0.1876s_0)$, we have

$$\begin{aligned} & \inf_{\hat{\eta} \in \{0,1\}^{d \times m}} \sup_{\beta^* \in \Theta_{g,1}} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\eta}(Y, X) - \eta^*\|_2^2 \\ & \geq \frac{(ss_0)^{39/40}}{6} \frac{\sqrt{2/\pi}}{1 + \sqrt{(\log(ss_0))/20}} - \frac{10}{3} ss_0 \exp \left(-\frac{ss_0}{14} \right) - 4s^2 s_0 \exp(-0.1876s_0) \\ & \geq \frac{1}{6} (ss_0)^{\frac{4}{5}} - 0.016(ss_0)^{\frac{4}{5}} - 0.14(ss_0)^{\frac{4}{5}} \end{aligned}$$

$$> \frac{1}{100} (s s_0)^{\frac{4}{5}}.$$

Therefore, we complete the proof of Theorem 3.3.

Appendix C: Auxiliary lemmas for the oracle properties

Firstly, we introduce a useful lemma from Theorem 2.1 in [17]. The conclusion of this lemma is not limited to the true sparsity level (s, s_0) ; in fact, it can be applied to any $0 < s' < m$ and $0 < s'_0 < d/e$.

Lemma C.1. *Assume that $X \in \mathbb{R}^{n \times p}$ satisfies $\text{DSRIP}(s', s'_0, \delta)$ with $\delta \in (0, 1)$. For every $k \in [n]$, assume that each ξ_k is independent sub-Gaussian random variable with zero mean and $\|\xi\|_{\psi_2}^2 \leq 2$. Define $\Xi_{ij} = \frac{\sigma}{n} X_{(ij)}^\top \xi$ and assume that $d \geq e s'_0$. Then the event*

$$\mathcal{E}(s', s'_0) := \left\{ \text{For every } S \in \mathcal{S}(s', s'_0), \sum_{(i,j) \in S} \Xi_{ij}^2 < \frac{10\sigma^2 s' s'_0 \Delta(s', s'_0)}{n} \right\}$$

holds with probability greater than $1 - \exp\left(-\frac{1}{3} s' s'_0 \Delta(s', s'_0)\right)$, where

$$\Delta(s', s'_0) := \frac{1}{s'_0} \log \frac{em}{s'} + \log \frac{ed}{s'_0}.$$

Proof of Lemma C.1. From Theorem 2.1 in [17], for all $t > 0$ and all $S \in \mathcal{S}(s', s'_0)$, we have

$$P\left(\|X_S^\top \xi\|_2^2 \geq \text{tr}(X_S X_S^\top) + 2\sqrt{\text{tr}(X_S X_S^\top X_S X_S^\top)t} + 2\|X_S X_S^\top\|_2 t\right) \leq e^{-t}, \quad (66)$$

where we also use $\|\cdot\|_2$ to denote the spectral norm of a matrix. Based on $\text{DSRIP}(s, s_0, \delta)$ condition, we have $n(1 - \delta) \leq \|X_S^\top X_S\|_2 \leq n(1 + \delta)$, which leads

$$\begin{aligned} \text{tr}(X_S X_S^\top) &= \text{tr}(X_S^\top X_S) \leq 2s' s'_0 n, \\ \text{tr}(X_S X_S^\top X_S X_S^\top) &= \text{tr}(X_S^\top X_S \cdot X_S^\top X_S) \leq |S| \cdot \|X_S^\top X_S\|_2^2 \leq 4n^2 s' s'_0, \\ \|X_S X_S^\top\|_2 &= \|X_S^\top X_S\|_2 \leq 2n. \end{aligned}$$

Back to (66), we have

$$P\left(\|X_S^\top \xi\|_2^2 \geq 2s' s'_0 n + 4n\sqrt{s' s'_0 t} + 4nt\right) \leq e^{-t}. \quad (67)$$

Then take $t = \frac{4}{3} s' s'_0 \Delta(s', s'_0)$, we obtain

$$2s' s'_0 n + 4n\sqrt{s' s'_0 t} + 4nt < 10ns' s'_0 \Delta(s', s'_0).$$

Therefore, for a fixed $S \in \mathcal{S}(s', s'_0)$ satisfying $|S| = s' s'_0$, we have

$$P\left(\sum_{(i,j) \in S} \Xi_{ij}^2 \geq \frac{10\sigma^2}{n} s' s'_0 \Delta(s', s'_0)\right) \leq \exp\left(-\frac{4}{3} s' s'_0 \Delta(s', s'_0)\right).$$

Finally, by the probability union bound, we have

$$\begin{aligned}
& P \left(\bigcup_{S \in \mathcal{S}(s', s'_0): |S|=s's'_0} \left\{ \sum_{(i,j) \in S} \Xi_{ij}^2 \geq \frac{10\sigma^2 s' s'_0 \Delta(s', s'_0)}{n} \right\} \right) \\
& \leq \binom{m}{s'} \binom{ds'}{s' s'_0} \exp \left(-\frac{4}{3} s' s'_0 \Delta(s', s'_0) \right) \\
& \leq \exp \left(-\frac{1}{3} s' s'_0 \Delta(s', s'_0) \right),
\end{aligned}$$

where the last inequality follows from $\binom{y}{x} \leq (ey/x)^x$ for every $0 < x < y$. Thus, we complete the proof of Lemma C.1. $\square$

C.1. Lemma C.2 and its proof

Now we focus on some inequalities used in the proof of Proposition 2.1. Specifically, the following lemma bounds the squared sum of $\tilde{\Xi}$ on some $S_{OG} \in \mathcal{S}(As, s_0)$. Recall that $\tilde{\Xi} := \frac{\sigma}{n} X^\top (\mathbf{I}_n - X_{S^*} (X_{S^*}^\top X_{S^*})^{-1} X_{S^*}^\top) \xi \in \mathbb{R}^p$, $A := \frac{8\delta^2}{(\kappa-\delta)^2}$, $\mu' := \sigma \sqrt{\frac{10}{n} [\Delta(1, s_0) + \log(ss_0)]}$, and $\mu \geq \sqrt{4 + \frac{12\delta^2}{(1-\delta)^2}} \mu'$.

Lemma C.2. *Under conditions of Proposition 2.1, for every falsely discovered set $S_{OG} \in \mathcal{S}(As, s_0)$ which does not contain any support group, with a probability greater than $1 - \exp\left(-\frac{1}{3}s_0\Delta(1, s_0)\right)$, we have*

$$\sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \leq \frac{(1-\delta)^2}{3} \|\tilde{\beta}^t - \tilde{\beta}^*\|_2^2.$$

Proof of Lemma C.2. Based on the definition of the double sparse operator $\mathcal{T}_{\mu, s_0}$ and $\mathcal{T}_{\mu', s_0}$, we have

$$\begin{aligned}
& \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \\
& = \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu', s_0}(\tilde{\Xi})_{ij} \neq 0, \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \\
& \quad + \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu', s_0}(\tilde{\Xi})_{ij} = 0, \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \\
& \leq \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{\mathcal{T}_{\mu', s_0}(\tilde{\Xi})_{ij} \neq 0\} + \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| < \mu', \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \\
& \quad + \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\left\{|\tilde{\Xi}_{ij}| \geq \mu', \sum_{k \in [d]} \tilde{\Xi}_{kj}^2 \mathbf{1}\{|\tilde{\Xi}_{kj}| \geq \mu'\} < s_0 \mu'^2, \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0\right\}.
\end{aligned}$$

Then, by combining Lemma C.3, C.4 and C.5 together, we complete the proof of Lemma C.2. $\square$

Lemma C.3. *Under all conditions of Proposition 2.1, we have*

$$P\left(\sum_{(i,j) \in [d] \times [m]} \mathbf{1}\{\mathcal{T}_{\mu',s_0}(\tilde{\Xi})_{ij} \neq 0\} = 0\right) \geq 1 - \exp\left(-\frac{1}{3}s_0\Delta(1, s_0)\right),$$

that is, no element or group of $\tilde{\Xi}$ can be selected by the operator $\mathcal{T}_{\mu',s_0} = \mathcal{T}_{\mu',s_0}^{(2)} \circ \mathcal{T}_{\mu'}^{(1)}$.

Proof of Lemma C.3. Firstly note that $\|\tilde{\Xi}_{ij}\|_{\psi_2}^2 \leq \frac{2\sigma^2}{n}$. Then we take $s' = 1$, $s'_0 = s_0$ and conclude that event $\mathcal{E}(1, s_0)$ (defined in Lemma C.1) holds with probability greater than $1 - \exp\left(-\frac{1}{3}s_0\Delta(1, s_0)\right)$. Based on $\mathcal{E}(1, s_0)$, we next prove that no group in $\tilde{\Xi}$ can be discovered under $\mathcal{T}_{\mu',s_0}$ by using contradiction. Specifically, if there exists a group G_{j_0} satisfies $\mathcal{T}_{\mu',s_0}(\tilde{\Xi}_{G_{j_0}}) \neq \mathbf{0}_d$, we can separate the proof into two cases:

1. If more than s_0 entries in $\tilde{\Xi}_{G_{j_0}}$ are discovered, we can select arbitrary s_0 discovered entries from $\tilde{\Xi}_{G_{j_0}}$, and use S' to denote their index set. Then, by the definition of the element-wise thresholding operator $\mathcal{T}_{\mu'}^{(1)}$ we obtain

$$\sum_{(i,j) \in S'} \tilde{\Xi}_{ij}^2 \geq s_0\mu'^2 > \frac{10\sigma^2 s_0\Delta(1, s_0)}{n},$$

which contradicts the event $\mathcal{E}(1, s_0)$.

2. If only less than s_0 entries in $\tilde{\Xi}_{G_{j_0}}$ is discovered, we select all discovered entries from $\tilde{\Xi}_{G_{j_0}}$, and use S'' to denote their index set. Then, by the definition of the group-wise thresholding operator $\mathcal{T}_{\mu',s_0}^{(2)}$ we obtain

$$\sum_{(i,j) \in S''} \tilde{\Xi}_{ij}^2 \geq s_0\mu'^2 > \frac{10\sigma^2 s_0\Delta(1, s_0)}{n},$$

which contradicts the event $\mathcal{E}(1, s_0)$ again.

Therefore, we prove that no group in $\tilde{\Xi}$ can be selected by the operator $\mathcal{T}_{\mu',s_0}$, which completes the proof of Lemma C.3. $\square$

Lemma C.4. *Under conditions of Proposition 2.1, for every falsely discovered set $S_{OG} \in \mathcal{S}(As, s_0)$ which does not contain any support group, we have*

$$\sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| < \mu', \mathcal{T}_{\mu,s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \leq \frac{(1-\delta)^2}{6} \|\tilde{\beta}^* - \tilde{\beta}^t\|_2^2. \quad (68)$$

Proof of Lemma C.4. By the decomposition $\tilde{H}_{ij}^{t+1} = \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle + \tilde{\Xi}_{ij}$ for every $(i, j) \in (S^*)^c$ and the DSRIP $\left((1+2A)s, \frac{1+4A}{1+2A}s_0, \delta\right)$ condition, we have

$$\begin{aligned} & \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| < \mu', \mathcal{T}_{\mu,s_0}(\tilde{H}^{t+1})_{ij} \neq 0\} \\ & \leq \sum_{(i,j) \in S_{OG}} \mu'^2 \mathbf{1}\{|\tilde{\Xi}_{ij}| < \mu', |\tilde{\Xi}_{ij} + \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu\} \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{(i,j) \in S_{OG}} \mu'^2 \mathbf{1} \left\{ |\tilde{\Xi}_{ij}| < \mu', |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu - \mu' \right\} \\
&\stackrel{(i)}{\leq} \sum_{(i,j) \in S_{OG}} \mu'^2 \mathbf{1} \left\{ \mu' \leq \frac{1-\delta}{\sqrt{6\delta}} |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \right\} \\
&\leq \frac{(1-\delta)^2}{6\delta^2} \sum_{(i,j) \in S_{OG}} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 \\
&\leq \frac{(1-\delta)^2}{6} \|\tilde{\beta}^* - \tilde{\beta}^t\|_2^2,
\end{aligned} \tag{69}$$

where inequality (i) follows from $\mu > \left(1 + \frac{\sqrt{6\delta}}{1-\delta}\right) \mu'$. Therefore, we complete the proof of Lemma C.4. $\square$

Lemma C.5. *Under conditions of Proposition 2.1, for every falsely discovered set $S_{OG} \in \mathcal{S}(As, s_0)$ which does not contain any support group, we use G_{OG} to represent the group index set of S_{OG} . Then, with a probability greater than $1 - \exp\left(-\frac{1}{3}s_0\Delta(1, s_0)\right)$, we have*

$$\begin{aligned}
&\sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1} \left\{ |\tilde{\Xi}_{ij}| \geq \mu', \sum_{k \in [d]} \tilde{\Xi}_{kj}^2 \mathbf{1}(|\tilde{\Xi}_{kj}| \geq \mu') < s_0 \mu'^2, \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0 \right\} \\
&\leq \frac{(1-\delta)^2}{6} \|\tilde{\beta}^t - \tilde{\beta}^*\|_2^2.
\end{aligned}$$

Proof of Lemma C.5. For any non-support group G_j satisfies that

$$\mathcal{T}_{\mu', s_0}(\tilde{\Xi}_{G_j}) = \mathbf{0}_d, \quad \mathcal{T}_{\mu, s_0}(\tilde{H}_{S_{OG} \cap G_j}^{t+1}) \neq \mathbf{0},$$

by the double sparse operator $\mathcal{T}_{\mu, s_0}$ and $\mu \geq \sqrt{4 + \frac{12\delta^2}{(1-\delta)^2}} \mu'$, we have

$$\begin{aligned}
s_0 \mu^2 &\leq \sum_{k:(k,j) \in S_{OG}} \underbrace{\left(\tilde{\Xi}_{kj} + \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle \right)^2}_{\tilde{H}_{kj}^{t+1}} \mathbf{1} \left(|\tilde{\Xi}_{kj} + \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu \right) \\
&\leq \sum_{k:(k,j) \in S_{OG}} 2 \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 \mathbf{1} \left(|\tilde{\Xi}_{kj} + \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu \right) \\
&\quad + \sum_{k:(k,j) \in S_{OG}} 2 \tilde{\Xi}_{kj}^2 \mathbf{1} \left(|\tilde{\Xi}_{kj} + \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \geq \mu \right) \\
&\leq 2 \sum_{k:(k,j) \in S_{OG}} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 + 2 \sum_{k:(k,j) \in S_{OG}} \tilde{\Xi}_{kj}^2 \mathbf{1} \left(|\tilde{\Xi}_{kj}| \geq \mu' \right) \\
&\quad + \sum_{k:(k,j) \in S_{OG}} 2 \tilde{\Xi}_{kj}^2 \mathbf{1} \left(|\tilde{\Xi}_{kj}| < \mu' \leq \frac{1-\delta}{\sqrt{6\delta}} |\langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \right) \\
&\leq 2s_0 \mu'^2 + 2 \left(1 + \frac{(1-\delta)^2}{6\delta^2} \right) \sum_{k:(k,j) \in S_{OG}} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2,
\end{aligned} \tag{70}$$

where the last inequality follows from

$$\sum_{k \in [d]} \tilde{\Xi}_{kj}^2 \mathbf{1}(|\tilde{\Xi}_{kj}| \geq \mu') < s_0 \mu'^2$$

by using the result in Lemma C.3. Therefore, based on (70), we conclude that

$$\frac{6\delta^2}{(1-\delta)^2} s_0 \mu'^2 \leq \sum_{k:(k,j) \in S_{OG}} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2.$$

Then, we get the upper bound as

$$\begin{aligned} & \sum_{(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1} \left\{ |\tilde{\Xi}_{ij}| \geq \mu', \sum_{k \in [d]} \tilde{\Xi}_{kj}^2 \mathbf{1}(|\tilde{\Xi}_{kj}| \geq \mu') < s_0 \mu'^2, \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0 \right\} \\ &= \sum_{j \in G_{OG}} \sum_{i:(i,j) \in S_{OG}} \tilde{\Xi}_{ij}^2 \mathbf{1} \left\{ |\tilde{\Xi}_{ij}| \geq \mu' \right\} \\ & \quad \times \mathbf{1} \left\{ \sum_{k \in [d]} \tilde{\Xi}_{kj}^2 \mathbf{1}(|\tilde{\Xi}_{kj}| \geq \mu') < s_0 \mu'^2, \mathcal{T}_{\mu, s_0}(\tilde{H}^{t+1})_{ij} \neq 0 \right\} \quad (71) \\ &\leq \sum_{j \in G_{OG}} s_0 \mu'^2 \mathbf{1} \left\{ s_0 \mu'^2 \leq \frac{(1-\delta)^2}{6\delta^2} \sum_{k:(k,j) \in S_{OG}} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 \right\} \\ &\leq \frac{(1-\delta)^2}{6\delta^2} \sum_{(i,j) \in S_{OG}} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 \\ &\leq \frac{(1-\delta)^2}{6} \|\tilde{\beta}^t - \tilde{\beta}^*\|_2^2. \end{aligned}$$

Therefore, we complete the proof of Lemma C.5. $\square$

C.2. Lemma C.6 and its proof

Finally, we analyze one essential error term associated with the true support set S^* .

Lemma C.6. *Under conditions of Proposition 2.1, recall that the element-wise and the group-wise signal conditions*

$$\begin{aligned} \min_{(i,j) \in S^*} |\beta_{ij}^*| &\geq \left(2 + \frac{\sqrt{6}\delta}{1-\delta}\right) \mu, \\ \min_{j \in G^*} \|\beta_{G_j}^*\|_2 &\geq \left(2 + \frac{\sqrt{6}\delta}{1-\delta}\right) \sqrt{s_0} \mu. \end{aligned}$$

Then, we have

$$P \left(\sum_{(i,j) \in S^*} \left(\tilde{H}_{ij}^{t+1} \right)^2 \mathbf{1} \{ (i,j) \notin \tilde{S}^{t+1} \} < \frac{(1-\delta)^2}{3} \|\tilde{\beta}^t - \tilde{\beta}^*\|_2^2 \right) \geq 1 - O \left(e^{-\frac{1}{3} [\Delta(1, s_0) + \log(s_0)]} \right).$$

Proof of Lemma C.6. Recall $s_j = |G_j \cap S^*|$ for all $j \in G^*$. Then we have

$$\sum_{(i,j) \in S^*} \left(\tilde{H}_{ij}^{t+1} \right)^2 \mathbf{1} \left((i,j) \notin \tilde{S}^{t+1} \right)$$

$$\begin{aligned}
&\leq \sum_{(i,j) \in S^*} \left(\tilde{H}_{ij}^{t+1} \right)^2 \mathbf{1} \left(|\tilde{H}_{ij}^{t+1}| < \mu \right) \\
&\quad + \sum_{(i,j) \in S^*} \left(\tilde{H}_{ij}^{t+1} \right)^2 \cdot \mathbf{1} \left(|\tilde{H}_{ij}^{t+1}| \geq \mu, \sum_{k \in [d]} \left(\tilde{H}_{kj}^{t+1} \right)^2 \mathbf{1} (|\tilde{H}_{kj}^{t+1}| \geq \mu) < s_0 \mu^2 \right) \\
&\leq \sum_{(i,j) \in S^*} \mu^2 \cdot \mathbf{1} \left(|\tilde{H}_{ij}^{t+1}| < \mu \right) + \sum_{j \in G^*} s_0 \mu^2 \cdot \mathbf{1} \left(\sum_{k: (k,j) \in S^*} \left(\tilde{H}_{kj}^{t+1} \right)^2 \mathbf{1} (|\tilde{H}_{kj}^{t+1}| \geq \mu) < s_0 \mu^2 \right) \\
&\leq \sum_{(i,j) \in S^*} \mu^2 \cdot \mathbf{1} \left(|\tilde{H}_{ij}^{t+1}| < \mu \right) + \sum_{j \in G^*} s_0 \mu^2 \cdot \mathbf{1} \left(\sum_{k: (k,j) \in S^*} \left(\tilde{H}_{kj}^{t+1} \right)^2 < (s_j + s_0) \mu^2 \right). \quad (72)
\end{aligned}$$

For all $(i, j) \in S^*$, we have $\tilde{H}_{ij}^{t+1} = \tilde{\beta}^* + \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle$ holds (since $\tilde{\Xi}_{ij} = 0$ for all $(i, j) \in S^*$). Therefore, for the first term in (72), we have

$$\begin{aligned}
\mathbf{1} \left(|\tilde{H}_{ij}^{t+1}| < \mu \right) &\leq \mathbf{1} \left(|\tilde{\beta}_{ij}^*| - |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| < \mu \right) \\
&\stackrel{(i)}{\leq} \mathbf{1} \left(|\beta_{ij}^*| - \frac{\mu}{3} - |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| < \mu \right) \\
&\leq \mathbf{1} \left(\mu < \frac{1-\delta}{\sqrt{6}\delta} |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \right), \quad (73)
\end{aligned}$$

where inequality (i) follows from the third inequality in (29), and the last inequality follows from the element-wise beta-min condition $\min_{(i,j) \in S^*} |\beta_{ij}^*| \geq \left(2 + \frac{\sqrt{6}\delta}{1-\delta}\right) \mu$.

Additionally, for the second term in (72), by both element-wise and group-wise beta-min conditions, we conclude that $\|\beta_{G_j}^*\|_2 \geq \left(2 + \frac{\sqrt{6}\delta}{1-\delta}\right) \sqrt{s_j \vee s_0} \mu$. Then, we have

$$\begin{aligned}
&\mathbf{1} \left(\sum_{k: (k,j) \in S^*} \left(\tilde{H}_{kj}^{t+1} \right)^2 < (s_j + s_0) \mu^2 \right) \\
&\leq \mathbf{1} \left(\sqrt{\sum_{k: (k,j) \in S^*} \left(\tilde{\beta}_{kj}^* \right)^2} - \sqrt{\sum_{k: (k,j) \in S^*} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2} < \sqrt{s_j + s_0} \mu \right) \\
&\leq \mathbf{1} \left(\sqrt{\sum_{k: (k,j) \in S^*} \left(\beta_{kj}^* \right)^2} - \sqrt{\sum_{k: (k,j) \in S^*} \left(\tilde{\beta}_{kj}^* - \beta_{kj}^* \right)^2} - \sqrt{2(s_j \vee s_0)} \mu \right. \\
&\quad \left. < \sqrt{\sum_{k: (k,j) \in S^*} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2} \right) \\
&\leq \mathbf{1} \left(\sum_{k: (k,j) \in S^*} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 > \frac{6\delta^2}{(1-\delta)^2} (s_j \vee s_0) \mu^2 > \frac{6\delta^2}{(1-\delta)^2} s_0 \mu^2 \right), \quad (74)
\end{aligned}$$

where the last inequality follows from the third inequality in (29).

By applying (73) and (74) into (72), we derive that

$$\sum_{(i,j) \in S^*} \left(\tilde{H}_{ij}^{t+1} \right)^2 \mathbf{1} \left((i, j) \notin \tilde{S}^{t+1} \right)$$

$$\begin{aligned}
&\leq \sum_{(i,j) \in S^*} \mu^2 \cdot \mathbf{1} \left(\mu < \frac{1-\delta}{\sqrt{6}\delta} |\langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle| \right) \\
&\quad + \sum_{j \in G^*} s_0 \mu^2 \cdot \mathbf{1} \left(\sum_{k: (k,j) \in S^*} \langle \Phi_{(k,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 > \frac{6\delta^2}{(1-\delta)^2} s_0 \mu^2 \right) \\
&\leq \frac{(1-\delta)^2}{3\delta^2} \sum_{(i,j) \in S^*} \langle \Phi_{(i,j)}^\top, \tilde{\beta}^* - \tilde{\beta}^t \rangle^2 \\
&\leq \frac{(1-\delta)^2}{3} \|\tilde{\beta}^* - \tilde{\beta}^t\|_2^2,
\end{aligned}$$

which completes the proof of Lemma C.6. $\square$

Appendix D: Auxiliary lemmas for the minimax lower bounds

This appendix provides some lemmas used in Appendix B.

Proof of Lemma B.1. To facilitate the calculation, we use the double index (i, j) to locate the i -th variable in the j -th group G_j . Recall $\pi_\Theta(A) = \frac{\pi(A)}{\pi(\Theta)}$. Then, based on the property of expectation, we have

$$\begin{aligned}
\inf_{\hat{\beta}} \sup_{\beta^* \in \Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\beta} - \beta^*\|_2^2 &\geq \inf_{\hat{\beta}} \mathbf{E}_{\beta^* \sim \pi_\Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\beta} - \beta^*\|_2^2 \\
&\geq \sum_{i=1}^d \sum_{j=1}^m \inf_{\hat{\beta}_{ij}} \mathbf{E}_{\beta^* \sim \pi_\Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \left(\hat{\beta}_{ij} - \beta_{ij}^{*\Theta} \right)^2 \\
&\stackrel{(i)}{=} \sum_{i,j} \inf_{\hat{\beta}_{ij}} \mathbf{E}_Y \left\{ \mathbf{E}_{\beta^{*\Theta}|Y} \left(\left(\hat{\beta}_{ij} - \beta_{ij}^{*\Theta} \right)^2 \middle| Y \right) \right\} \\
&\stackrel{(ii)}{=} \sum_{i,j} \mathbf{E}_Y \left\{ \mathbf{E}_{\beta^{*\Theta}|Y} \left(\left(B_{ij}^\Theta - \beta_{ij}^{*\Theta} \right)^2 \middle| Y \right) \right\} \\
&= \mathbf{E}_{\beta^* \sim \pi_\Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \left(\sum_{i,j} \left(B_{ij}^\Theta - \beta_{ij}^{*\Theta} \right)^2 \right),
\end{aligned} \tag{75}$$

where equality (i) uses $\mathbf{E}_{\beta^{*\Theta}|Y}$ to indicate that the expectation is with respect to $\beta^{*\Theta}$ conditioned on Y , that is, the conditional distribution $\frac{P(Y|\beta^*)d\pi_\Theta(\beta^*)}{\int P(Y|\beta)d\pi_\Theta(\beta^*)}$. And in inequality (ii) we define $B_{ij}^\Theta := \mathbf{E}_{\beta^{*\Theta}|Y}(\beta_{ij}^\Theta|Y)$, where recall that the distribution of β_{ij}^Θ is independently identical with that of $\beta_{ij}^{*\Theta}$. According to Theorem 1.1 and Corollary 1.2 in Chapter 4 in [21], B_{ij}^Θ achieves the infimum. Additionally, we have $(B_{ij}^\Theta)^2 \leq \mathbf{E}_{\beta^{*\Theta}|Y} \left(\left(\beta_{ij}^\Theta \right)^2 \middle| Y \right)$.

Next, by taking the infimum over all possible estimator $\hat{T}(Y, X)$, we obtain

$$\begin{aligned}
&\inf_{\hat{T}} \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}(Y, X) - \beta^*\|_2^2 \\
&\leq \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \sum_{i,j} \left(B_{ij}^\Theta - \beta_{ij}^* \right)^2
\end{aligned}$$

$$\begin{aligned}
&\leq \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \left(\sum_{i,j} \left(B_{ij}^\Theta - \beta_{ij}^* \right)^2 \mathbf{1}(\beta^* \in \Theta) \right) + \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \left(\sum_{i,j} \left(B_{ij}^\Theta - \beta_{ij}^* \right)^2 \mathbf{1}(\beta^* \notin \Theta) \right) \\
&\leq \mathbf{E}_{\beta^* \sim \pi_\Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \left(\sum_{i,j} \left(B_{ij}^\Theta - \beta_{ij}^* \right)^2 \right) \\
&\quad + 2 \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \sum_{i,j} \left(\mathbf{E}_{\beta^\Theta | Y} \left(\left(\beta_{ij}^\Theta \right)^2 | Y \right) + \left(\beta_{ij}^* \right)^2 \right) \mathbf{1}(\beta^* \notin \Theta) \right\}. \tag{76}
\end{aligned}$$

Combining (76) and (75) together, we get the lower bound of the minimax risk, that is,

$$\begin{aligned}
&\inf_{\hat{\beta}} \sup_{\beta^* \in \Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{\beta} - \beta^*\|_2^2 \\
&\geq \mathbf{E}_{\beta^* \sim \pi_\Theta} \mathbf{E}_{Y \sim P_{\beta^*}} \left(\sum_{i,j} \left(B_{ij}^\Theta - \beta_{ij}^\Theta \right)^2 \right) \\
&\geq \inf_{\hat{T}} \left\{ \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \|\hat{T}(Y, X) - \beta^*\|_2^2 \right\} - 2 \mathbf{E}_{\beta^* \sim \pi} \mathbf{E}_{Y \sim P_{\beta^*}} \left\{ \left(\mathbf{E}_{\beta^\Theta | Y} \left(\|\beta^\Theta\|_2^2 | Y \right) + \|\beta^*\|_2^2 \right) \mathbf{1}(\beta^* \notin \Theta) \right\},
\end{aligned}$$

which completes the proof of Lemma B.1. $\square$

The next lemma is a concentration inequality of the binomial distribution in Appendix D.4 in [28].

Lemma D.1 (Bernstein's inequality). *Suppose $u \sim \text{Bin}(n, p)$, $p \in (0, 1)$, then, for every $\lambda > 0$,*

$$P\left(\frac{u}{n} - p \geq \lambda\right) \leq \exp\left(-\frac{n\lambda^2}{2p(1-p) + \frac{2\lambda}{3}}\right).$$

The following lemma demonstrates the monotonicity of a function, which will be utilized in the proof of Theorem 3.1.

Lemma D.2. *Recall that*

$$\begin{aligned}
t(d, s_0, a, \sigma) &:= \frac{a\sqrt{n}}{2} + \frac{\sigma^2}{a\sqrt{n}} \log \frac{d - s_0}{s_0}, \\
\psi(d, s_0, a, \sigma) &:= (d - s_0) \Phi\left(-\frac{t(d, s_0, a, \sigma)}{\sigma}\right) + s_0 \Phi\left(-\frac{a\sqrt{n} - t(d, s_0, a, \sigma)}{\sigma}\right).
\end{aligned}$$

Then for the fixed d, s_0, a, σ , we have

$$\frac{\psi(d, s'_0, a, \sigma)}{s'_0} \geq \frac{\psi(d, s_0, a, \sigma)}{s_0}, \text{ for every } s'_0 \in (0, s_0].$$

Proof of Lemma D.2. For ease of display, for the fixed d, a, σ and for every $r \in (0, d)$, define

$$A(r) := \frac{\psi(d, r, a, \sigma)}{r} = \frac{d - r}{r} \Phi\left(-\frac{t(d, r, a, \sigma)}{\sigma}\right) + \Phi\left(-\frac{a\sqrt{n} - t(d, r, a, \sigma)}{\sigma}\right).$$

Thus we have

$$A'(r) = -\frac{d}{r^2} \Phi\left(-\frac{t}{\sigma}\right) + \frac{1}{\sigma} \frac{\partial t}{\partial r} \cdot \left\{ -\frac{d - r}{r} \varphi\left(-\frac{t}{\sigma}\right) + \varphi\left(-\frac{a\sqrt{n} - t}{\sigma}\right) \right\},$$

where $\varphi(\cdot)$ is the probability density function of the standard Gaussian distribution.

Note that

$$\begin{aligned}\frac{d-r}{r}\varphi\left(-\frac{t}{\sigma}\right) &= \frac{1}{\sqrt{2\pi}} \frac{d-r}{r} \exp\left(-\frac{1}{2\sigma^2} \left[\frac{a^2n}{4} + \frac{\sigma^4}{a^2n} \log^2 \frac{d-r}{r} + \sigma^2 \log \frac{d-r}{r}\right]\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2} \left[\frac{a^2n}{4} + \frac{\sigma^4}{a^2n} \log^2 \frac{d-r}{r} - \sigma^2 \log \frac{d-r}{r}\right]\right) \\ &= \varphi\left(-\frac{a\sqrt{n}-t}{\sigma}\right),\end{aligned}$$

which leads $A'(r) = -\frac{d}{r^2}\Phi\left(-\frac{t}{\sigma}\right) < 0$ for all $r \in (0, d)$. Therefore, we complete the proof of Lemma D.2. $\square$

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